

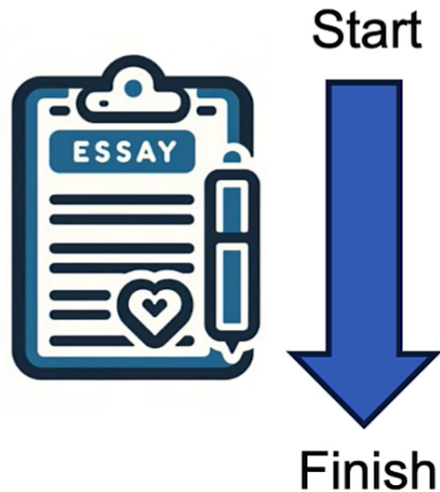
Agentic AI



Agentic AI

Non-agentic workflow (zero-shot):

Please type out an essay on topic X from start to finish in one go, without using backspace.



Agentic workflow:

Write an essay outline on topic X

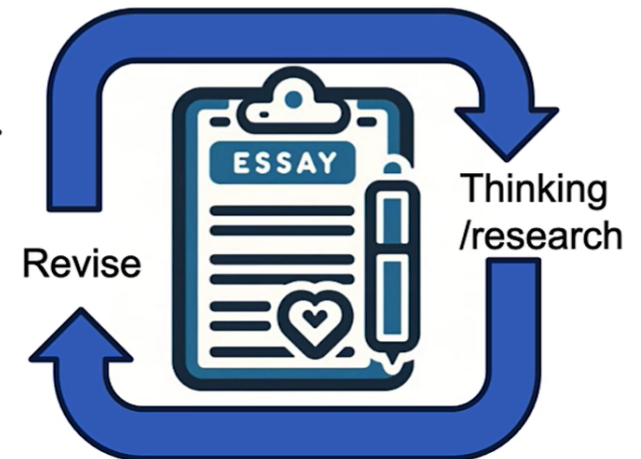
Do you need any web research?

Write a first draft.

Consider what parts need revision or more research.

Revise your draft.

....



Agentic AI workflows

An agentic AI workflow is a process where an LLM-based app executes multiple steps to complete a task.

Essay-writing example:

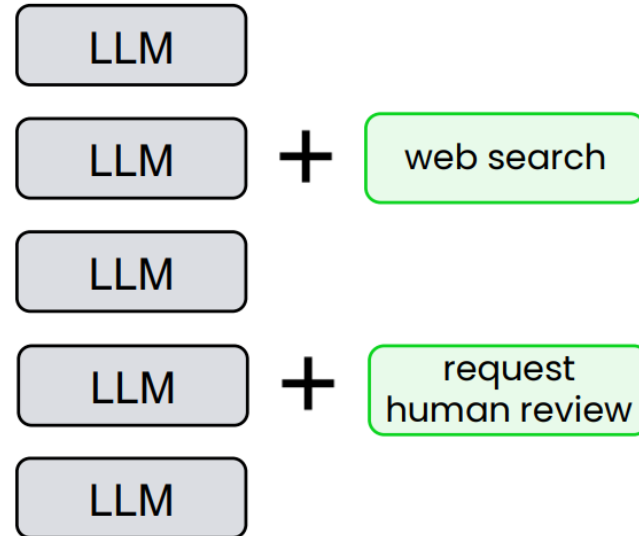
Write an essay outline on topic X

Do you need any web research?

Write a first draft.

Consider what parts need revision or more research.

Revise your draft.



It's an Agent or not?



Sequoia Ascent, March 2024



....

Rather than arguing over which work to include or exclude as being a true agent, we can acknowledge that there are different degrees to which systems can be agentic.

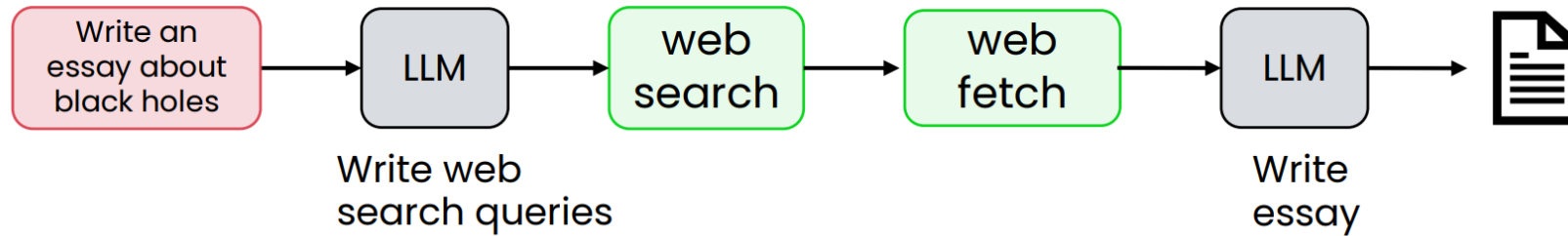
....



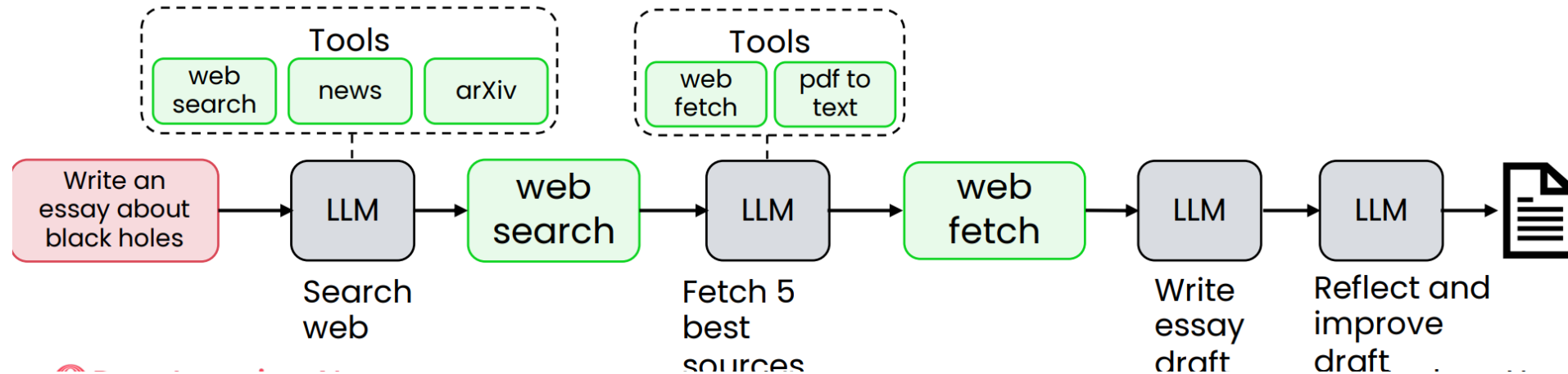
X (twitter) post , June 2024

Degrees of autonomy

Less autonomous



More autonomous



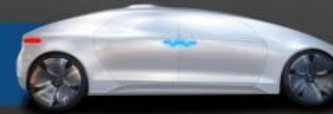
Degrees of autonomy



Levels of autonomy

Level 5

full autonomy,
no driver needed



Level 4

the car drives itself
in almost all situations
without the driver's help



Pictures courtesy of: <https://waymo.com/press>

Level 3

the driver doesn't have to
keep their eyes on the road
in certain situations



Level 2

the car accelerates, steers
and breaks on its own
monitored by the driver



Level 1

Basic assistance
(ABS, ESP, cruise
control, etc.)



Level 0

no autonomy,
the driver has
full control



Degrees of “agenticness”

Agentic AI can be less or more autonomous

Less
autonomous

Semi-
autonomous

Highly
autonomous



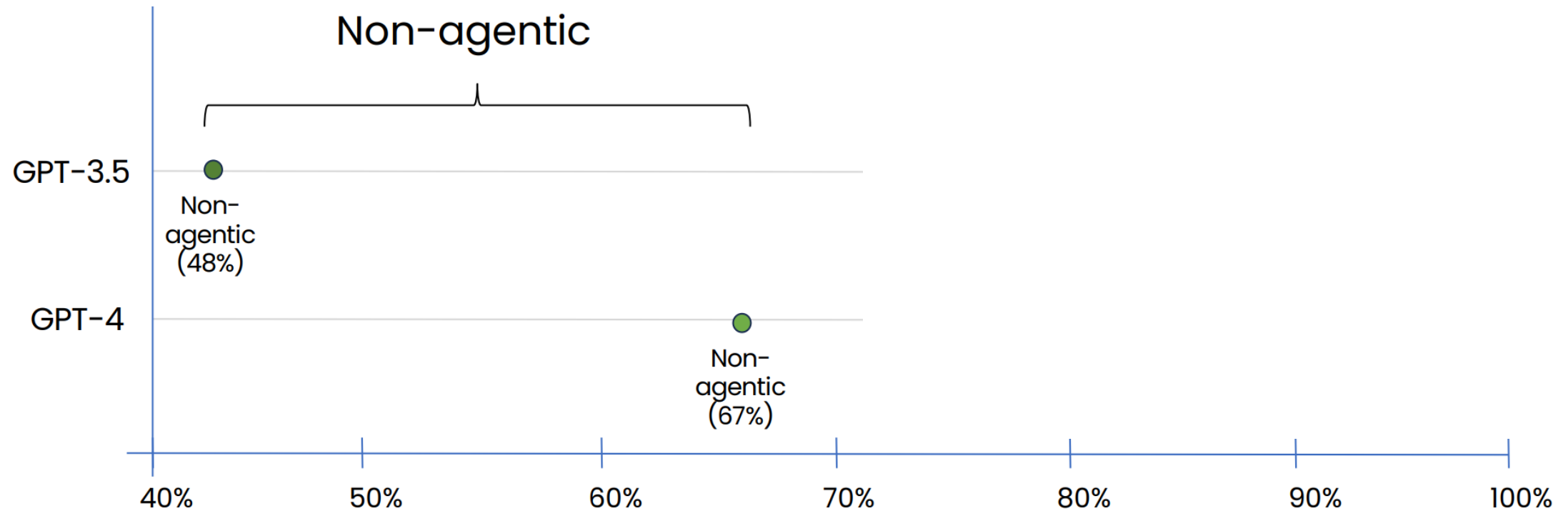
- All steps predetermined
- All tool use hard coded
- Autonomy is in text generation

- Agent can make some decisions, choose tools
- All tools predefined

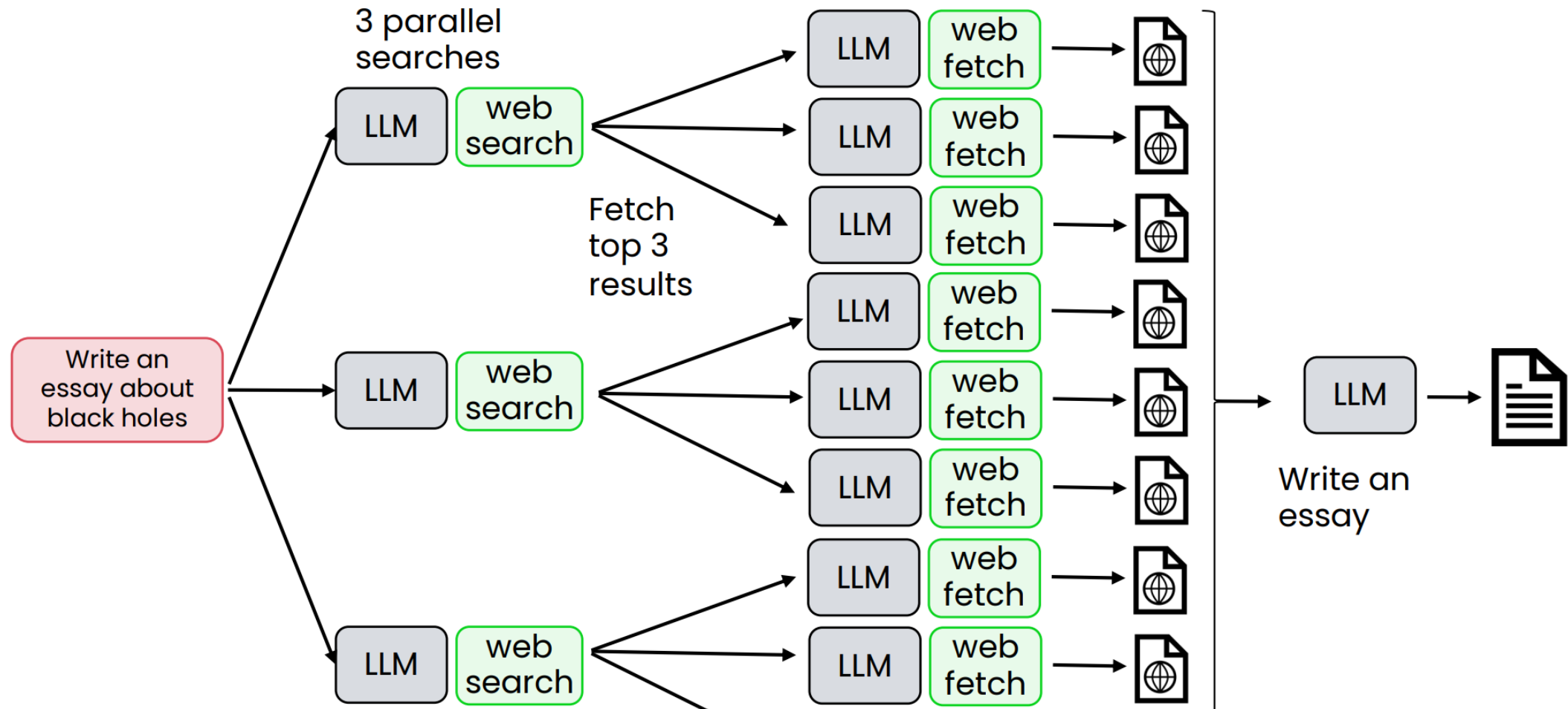
- Agent makes many decisions autonomously
- Can create new tools on the fly

Benefits of agentic AI

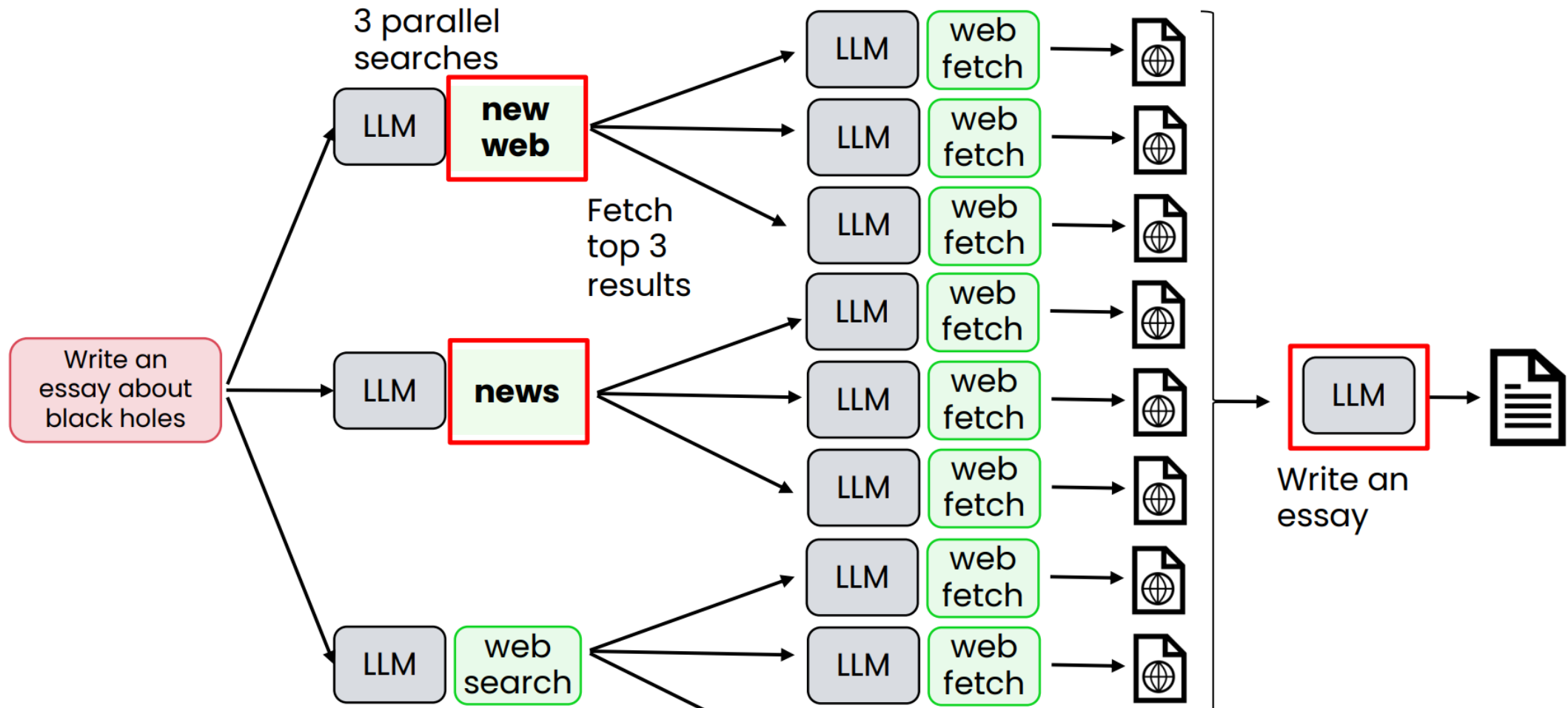
- Coding benchmark



Parallelization



Modularity



Key benefits of agentic workflows

- Performance
- Parallelization
- Modularity

Example: Invoice processing workflow

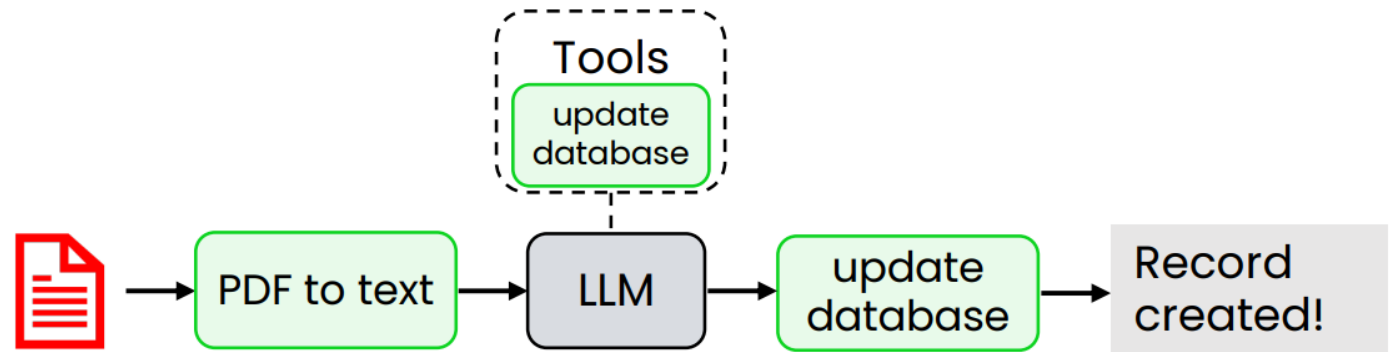
TechFlow Solutions LLC 890 Juniper Drive San Mateo, CA 94401 Phone: (415) 555-7890 Email: billing@techflowsol.com			
Due Date: August 20, 2025		Invoice Date: August 6, 2025	
Description	Qty	Unit Price	Line Total
Consulting - Systems Integration (hrs)	20	\$150.00	\$3,000.00
Total Due:			\$3,000.00

4 required fields:

Biller
Biller address
Amount due
Due date

Steps:

1. Identify required fields
2. Record in database



Example: Responding to customer email

From: Susan Jones
Subject: Wrong item shipped

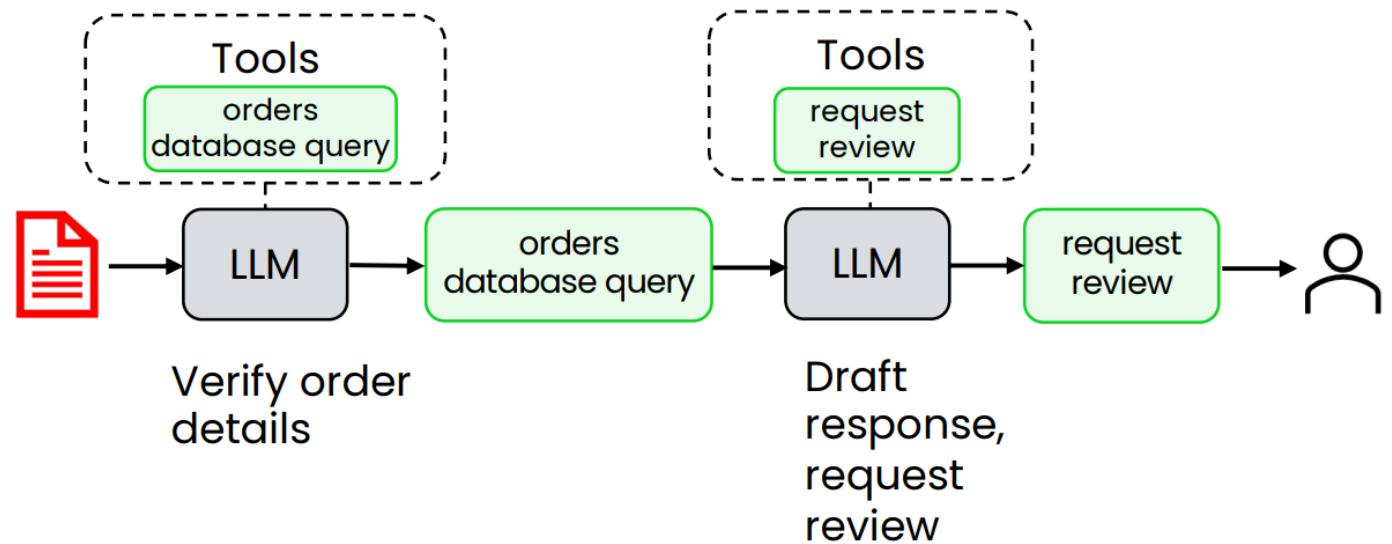
I ordered a blue KitchenPro blender (Order #8847) but received a red toaster instead.

I need the blender for my daughter's birthday party this weekend. Can you help?

Susan

Steps:

1. Extract key information
2. Find relevant customer records
3. Draft response for human review



More challenging: Customer service agent

Do you have any black jeans or blue jeans?

1. Check inventory for black jeans
2. Check inventory for blue jeans
3. Respond to customer

I'd like to return the beach towel I bought

1. Verify customer purchase
2. Check return policies
3. If return allowed = "yes", then:
 - a. Issue return packing slip
 - b. Set database record to "return pending"

Required steps
not known ahead
of time

Difficult: Visual computer use

Booking.com

PHP  

List your property

Register

Sign in

 Stays

 Flights

 Flight + Hotel

 Car rentals

 Attractions

 Airport taxis

[Home](#) > [Singapore](#) > [Singapore](#) > Search results

Search

Destination/property name:

Check-in date

 Fri 17 Mar

Check-out date

 Sat 18 Mar

1-night stay

2 adults · 0 children · 1 ro...

☐ I'm travelling for work

Search

Filter by:

Your previous filters

☐ 5 stars

64

☐ 1 star

13

☐ Breakfast included

130

☐ Parking

197

Popular filters

☐ Breakfast included

130

☐ Free cancellation

329

☐ No prepayment

231

☐ Less than 1 km

67

Singapore: 349 properties found

Show on map

Sort by: Our top picks All types of places

Your results include some shared accommodations, such as dormitory beds. [Show private rooms only](#)



ST Signature Jalan Besar

Little India, Singapore

1.7 km from centre

Metro access

Early 2023 Deal

Cabin M, Twin, Shared Bathroom

2 single beds

Good 7.5

470 reviews

1 night, 2 adults

~~₹ 8,892~~ ₹ 5,691

+₹ 455 taxes and charges

See availability



Pan Pacific Singapore

Marina Bay, Singapore

0.8 km from centre

Metro access

Travel Sustainable property

Early 2023 Deal

Deluxe King or Twin Room

Multiple bed types

Free cancellation

You can cancel later, so lock in this great price today.

Very good 8.5

1,651 reviews

Location 9.3

1 night, 2 adults

~~₹ 24,100~~ ₹ 18,316

+₹ 3,625 taxes and charges

See availability

What tasks is agentic AI suited to?

Easier

Harder



Clear, step-by-step process

Steps not known ahead of time

Standard procedures to follow

Plan/solve as you go

Text assets only

Multimodal (sound, vision)

Task decomposition

Write an essay on topic X



1. Write an essay outline on topic X



2. Search web



3. ~~Write the essay~~ Write a first draft



4. Consider what parts need revision

5. Revise your draft



Example: Responding to customer email

From: sjones9@email.com
Subject: Wrong item shipped

Hi, I ordered a blue KitchenPro blender (Order #8847) but received a red toaster instead.

I need the blender for my daughter's birthday party this weekend. Can you help?

Susan Jones

1. Extract key information

LLM

2. Find relevant customer records

LLM

+

orders database
query

3. Write and send response

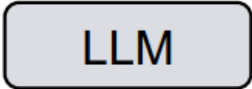
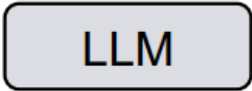
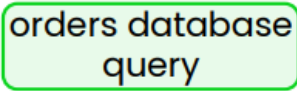
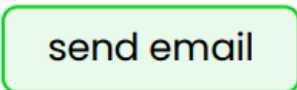
LLM

+

send email

Task decomposition

Example: Responding to email

1. Extract key information 
2. Find relevant customer records  + 
3. Write and send response  + 

Questions to ask:

- Can LLM, tool, or both complete each step?
- If “no”, consider breaking down into further steps, or adding human-in-the-loop

Building Blocks

Building block	Examples	Use cases
Models	LLMs	Text generation, tool use, information extraction
	Other AI models	PDF-to-text, text-to-speech, image analysis
Tools	API	Web search, get real-time data, send email, check calendar,....
	Information retrieval	Databases, Retrieval Augmented Generation (RAG)
	Code execution	Basic calculator, data analysis

Agentic design patterns

1. Reflection
2. Tool use
3. Planning
4. Multi-agent collaboration

Reflection

Reflection



Please write code for {task}

`def do_task(x): ...`

`def do_task_v2(x):`

`def do_task_v3(x):`



Coder
Agent
(LLM)

Here's code intended for {task}:

```
def do_task (x):  
    ...
```

Check the code carefully for correctness, style and efficiency, and give constructive criticism for how to improve it.

There's a bug on line 5. Fix it by ...

It failed Unit Test 3. Try changing ...

Reflection

Reflection



Please write code for {task}

`def do_task(x): ...`

`def do_task_v2(x):`

`def do_task_v3(x):`



Coder
Agent
(LLM)

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There's a bug on line 5. Fix it by ...

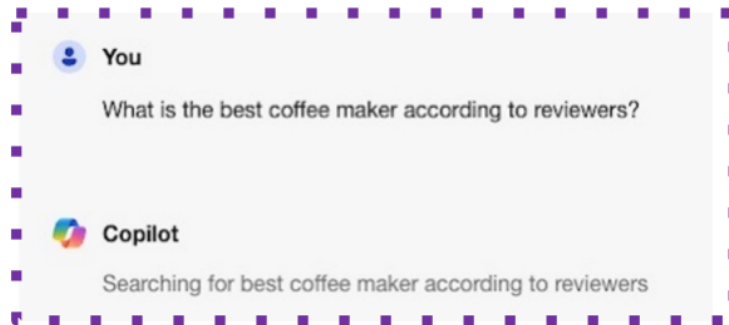
It failed Unit Test 3. Try changing ...



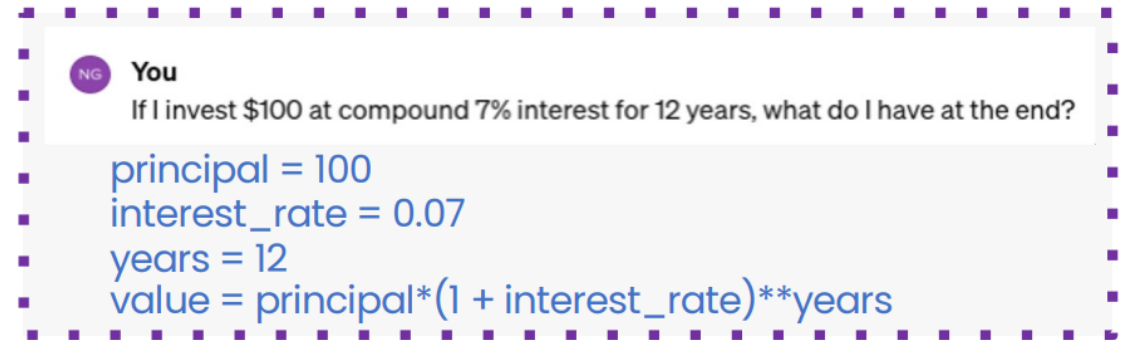
Critic
Agent
(LLM)

Tool use

Web search tool



Code execution tool



Analysis

- Code Execution
- Wolfram Alpha
- Bearly Code Interpreter

Information gathering

- Web search
- Wikipedia
- Database access

Productivity

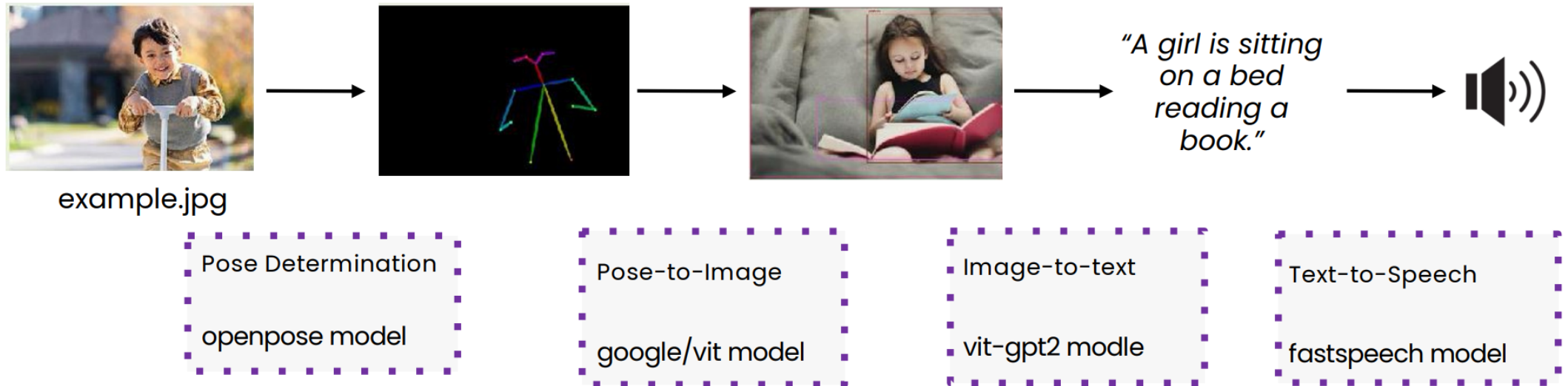
- Email
- Calendar
- Messaging

Images

- Image generation
- Image captioning
- OCR

Planning

Request: Please generate an image where a girl is reading a book, and her pose is the same as the boy in the image example.jpg, then please describe the new image with your voice.



[Example adapted from "HuggingGPT: Solving AI Tasks with ChatGPT and its Friends in Hugging Face", Shen et al. (2023)]

Multi-agentic workflows

ChatDev: A Zero-Code Multi-Agent Platform for Developing Everything



Multiagent Debate

Task	Single agent	Multi-agent
Biographies	66.0%	73.8%
MMLU	63.9%	71.1%
Chess move	29.3%	45.2%

[From "Improving Factuality and Reasoning in Language Models through Multiagent Debate", Du et al., 2023]

ChatDev

Launch

This is an article generation and polishing workflow. Please enter a word or a short phrase. The system will generate a full article based on your input and include a modern English poem. You can then provide revision suggestions, and the system will refine the article accordingly. You may give multiple rounds of feedback until you are satisfied.

17:21:47

Workflow execution started

17:21:55

17:21:55 User

Generate a New Year's Message

Poet 17:21:55

Model gpt-4o 00:00

00:00

Article Writer 17:21:55

Model gpt-4o 00:00

00:00

Please enter task prompt...

Upload File

WORKFLOW SELECTION

net_example_en.yaml

STATUS

Running...

VIEW

ChatGraph

Send

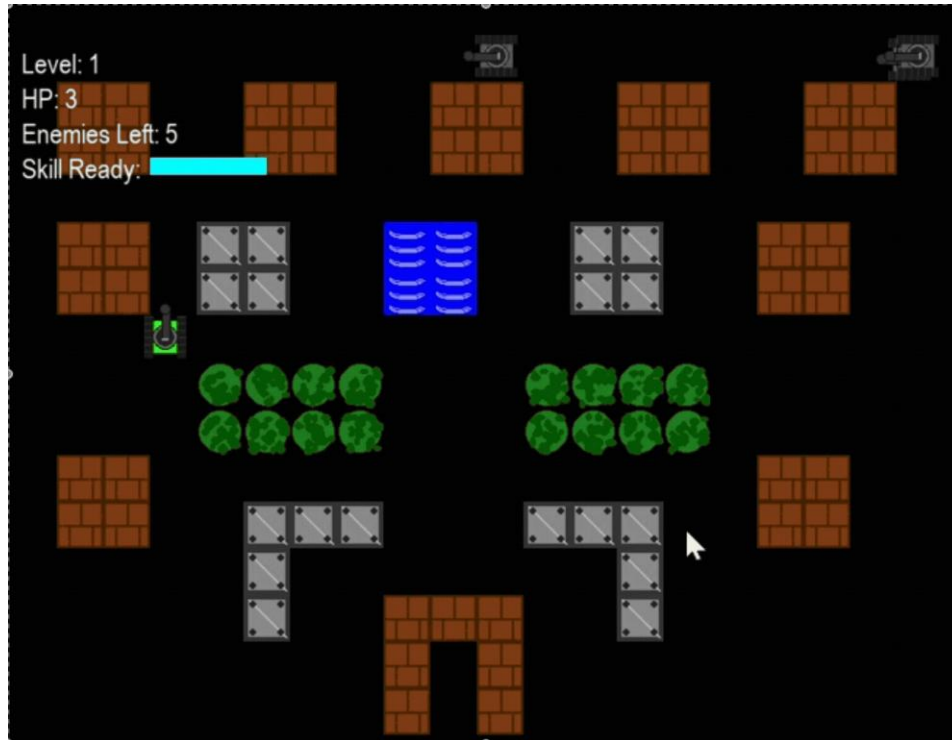
Cancel

Download Logs

ChatDev



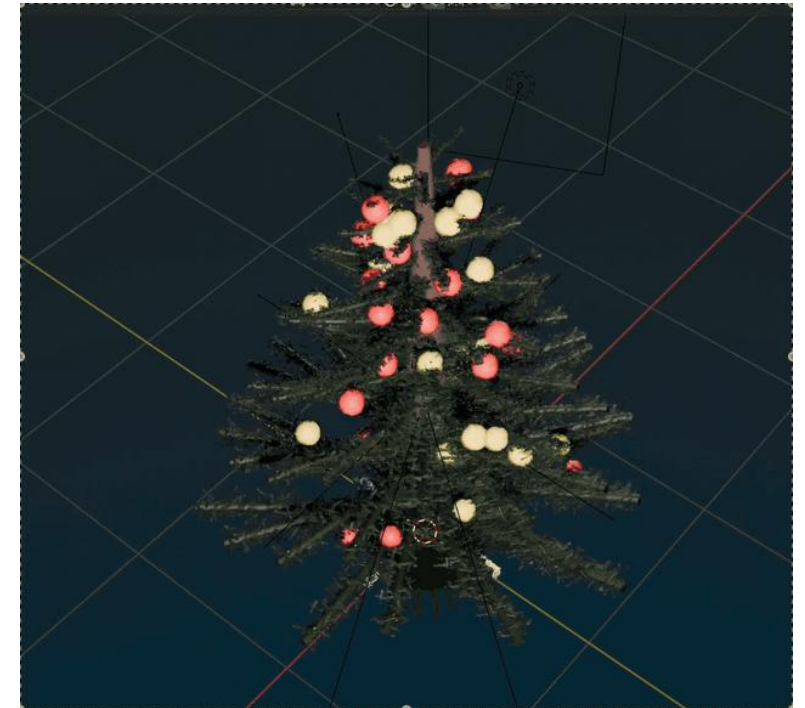
Game Dev



Prompt: *"Please help me design and develop a Tank Battle game."*



3D Generation



Prompt: *"Please build a Christmas tree."*

Reflection - humans



Write an
email



Didn't sign my
name

Hey Tommy,

I'll be in New York
next month, let me
know if you'll be
fre for dinner one
night.



Email V1

Specific
dates

Typo

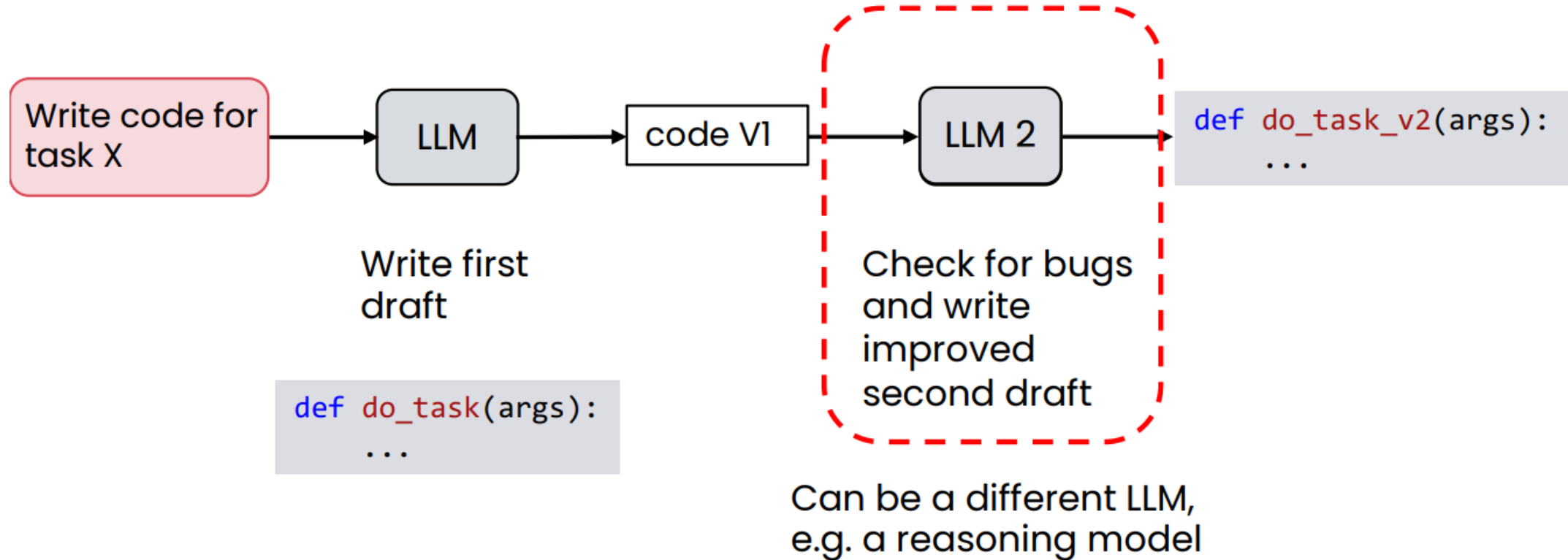
Hey Tommy,

I'll be in New York
next month from
the 5th–7th. Let me
know if you'll be
free for dinner one
night.

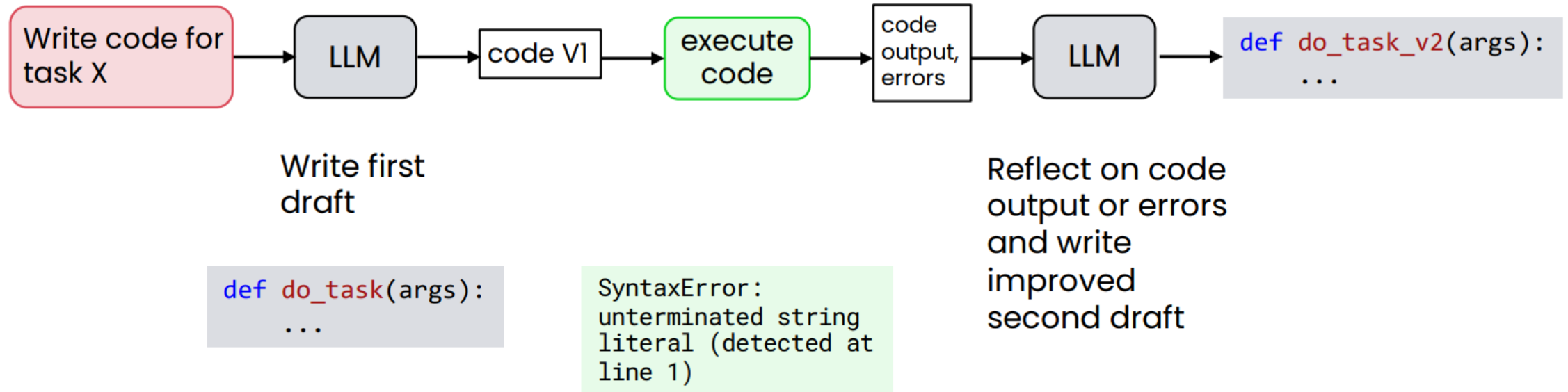
Andrew

Email V2

Reflection to improve code



Reflection with external feedback



SELF-REFINE paper

SELF-REFINE: Iterative Refinement with Self-Feedback

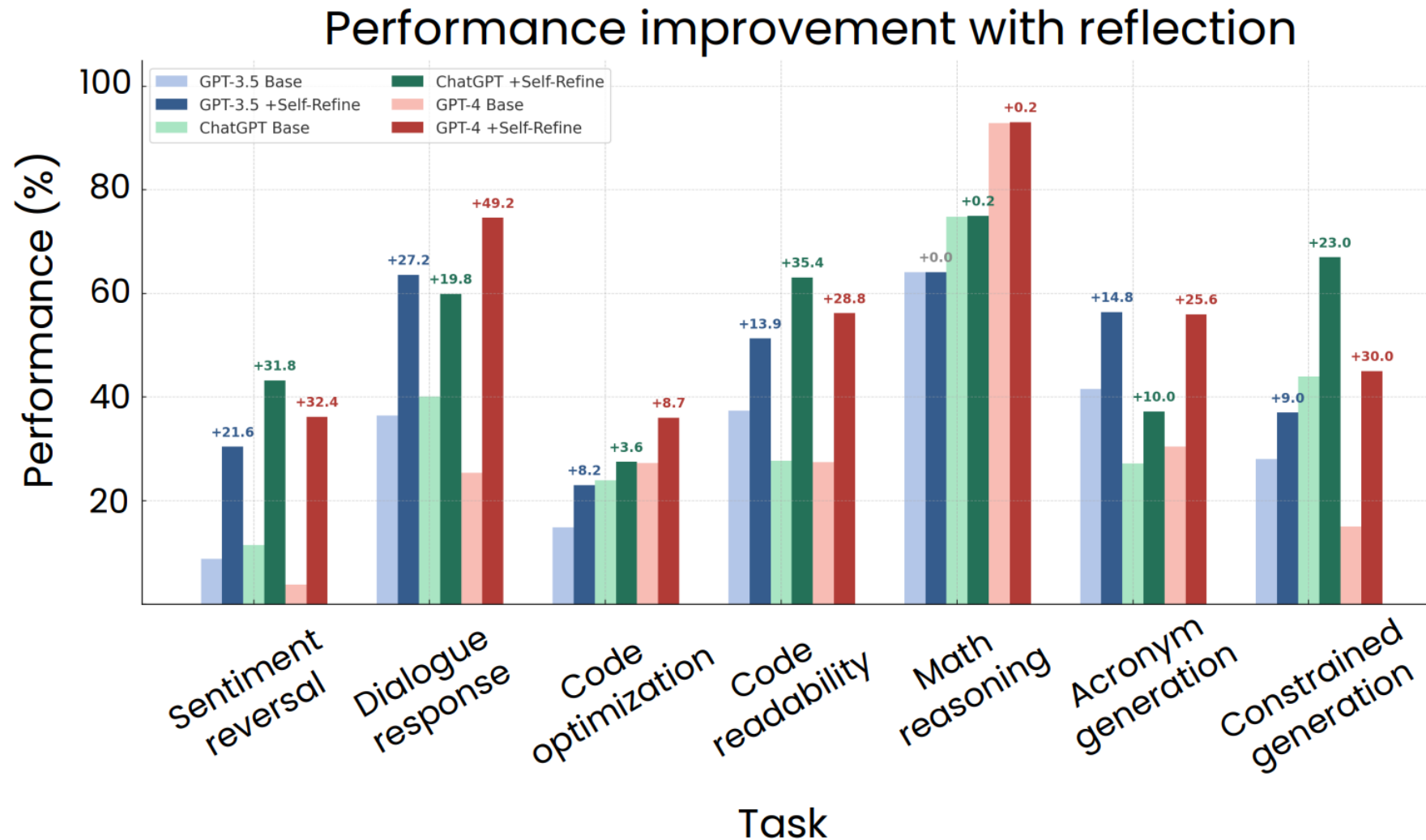
**Aman Madaan¹, Niket Tandon², Prakhar Gupta¹, Skyler Hallinan³, Luyu Gao¹,
Sarah Wiegrefe², Uri Alon¹, Nouha Dziri², Shrimai Prabhumoye⁴, Yiming Yang¹,
Shashank Gupta², Bodhisattwa Prasad Majumder⁵, Katherine Hermann⁶,
Sean Welleck^{2,3}, Amir Yazdanbakhsh⁶, Peter Clark²**

¹Language Technologies Institute, Carnegie Mellon University

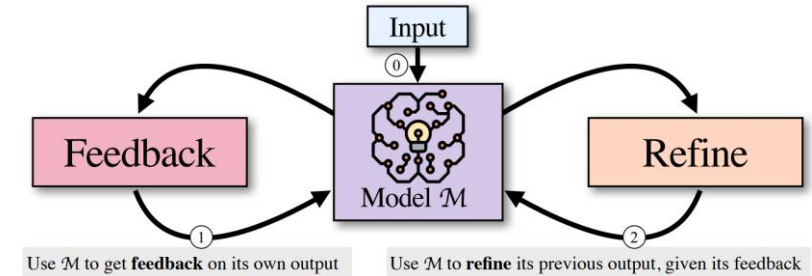
²Allen Institute for Artificial Intelligence

³University of Washington ⁴NVIDIA ⁵UC San Diego ⁶Google Research, Brain Team
amadaan@cs.cmu.edu, nikett@allenai.org

Reflection Results



Examples of SELF-REFINE



(a) Dialogue: x, y_t

User: I am interested in playing Table tennis.

Response: I'm sure it's a great way to socialize, stay active

(b) FEEDBACK fb

Engaging: Provides no information about table tennis or how to play it.

User understanding: Lacks understanding of user's needs and state of mind.

(c) REFINE y_{t+1}

Response (refined): That's great to hear (...) ! It's a fun sport requiring quick reflexes and good hand-eye coordination. Have you played before, or are you looking to learn?

(d) Code optimization: x, y_t

```
Generate sum of 1, ..., N
def sum(n):
    res = 0
    for i in range(n+1):
        res += i
    return res
```

(e) FEEDBACK fb

This code is slow as it uses brute force. A better approach is to use the formula ... $(n(n+1))/2$.

(f) REFINE y_{t+1}

Code (refined)

```
def sum_faster(n):
    return (n*(n+1))//2
```

Multiple iterations of FEEDBACK-REFINE

Task	y_0	y_1	y_2	y_3
Code Opt.	22.0	27.0	27.9	28.8
Sentiment Rev.	33.9	34.9	36.1	36.8
Constrained Gen.	29.0	40.3	46.7	49.7

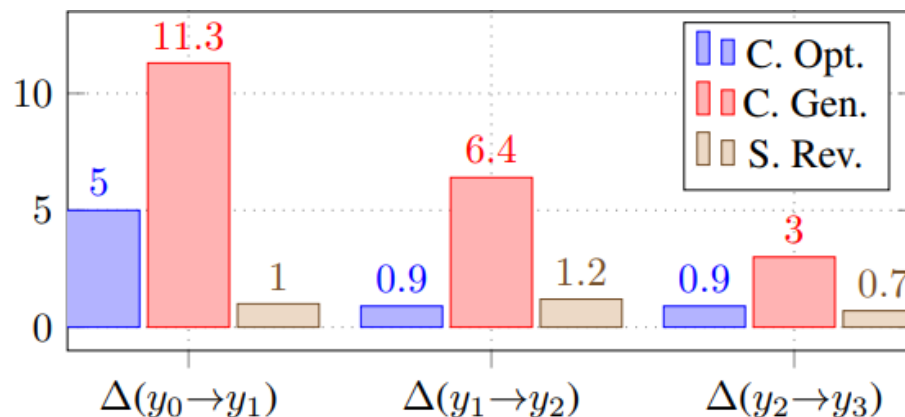
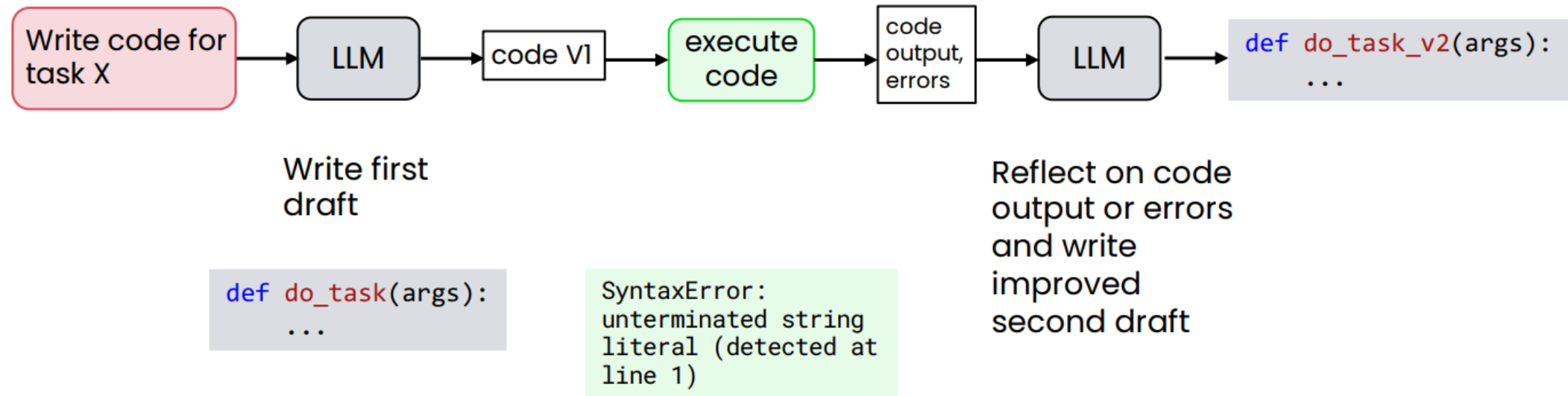


Figure 4: **Left:** Iteration-wise score improvements. Early iterations significantly improve output quality, and scores generally keep improving with more iterations. **Right:** SELF-REFINE Performance improvements with iterations. Most gains(Δ) are in the initial iterations for both Code Opt. and Sentiment Reversal. The numbers are averaged over ChatGPT, GPT-3.5, and GPT-4. Task abbreviations: C. Opt. (Code Optimiz.), S. Rev. (Sentiment Reversal), C. Gen. (Constrained Generation).

Other aspects

- Can we just generate multiple outputs instead of refining?
- Does SELF-REFINE work with weaker models?
- Etc.

Reflection with external feedback



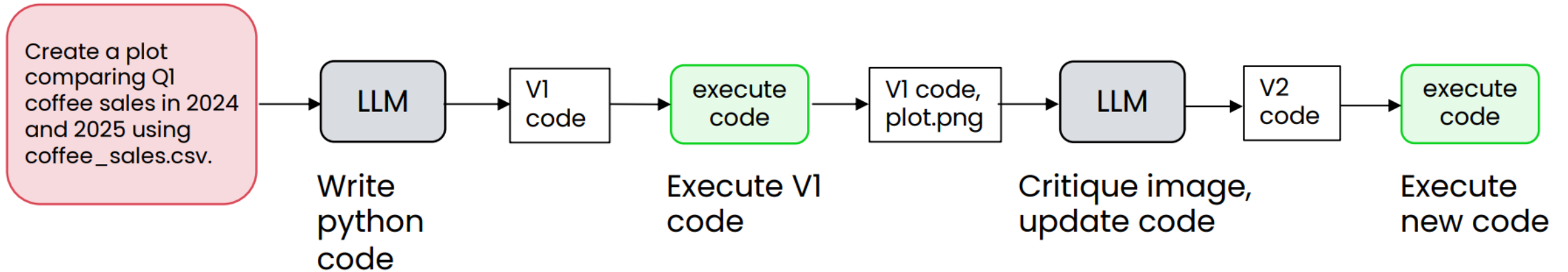
Example: Visualizing coffee sales



date	price	coffee_name
2024-01-12	3.87	Latte
2024-01-28	3.87	Hot Chocolate
2024-02-09	3.87	Hot Chocolate
2024-03-01	2.89	Cappuccino
2024-03-04	3.87	Latte
...	...	...
2025-03-23	3.57	Latte

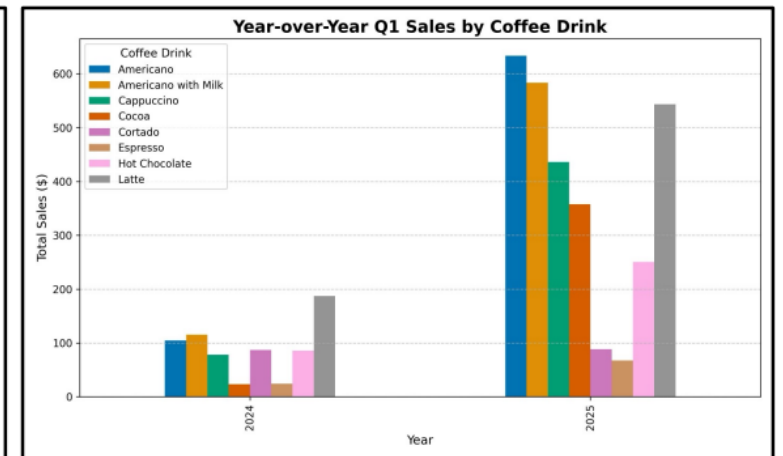
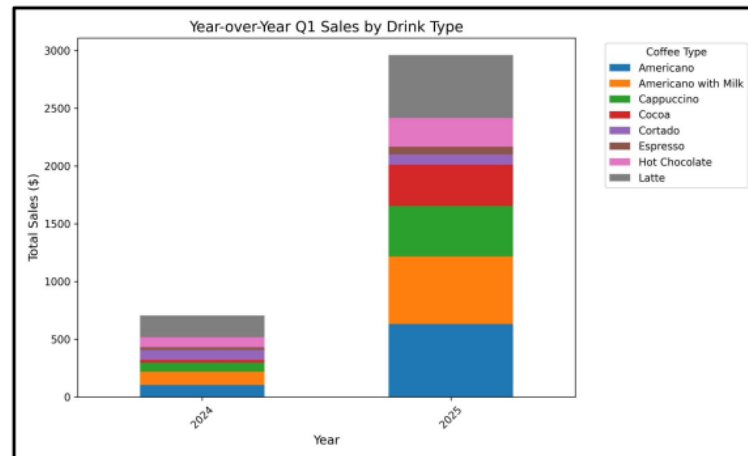
Create a plot comparing Q1 coffee sales in 2024 and 2025

Chart generation agentic workflow

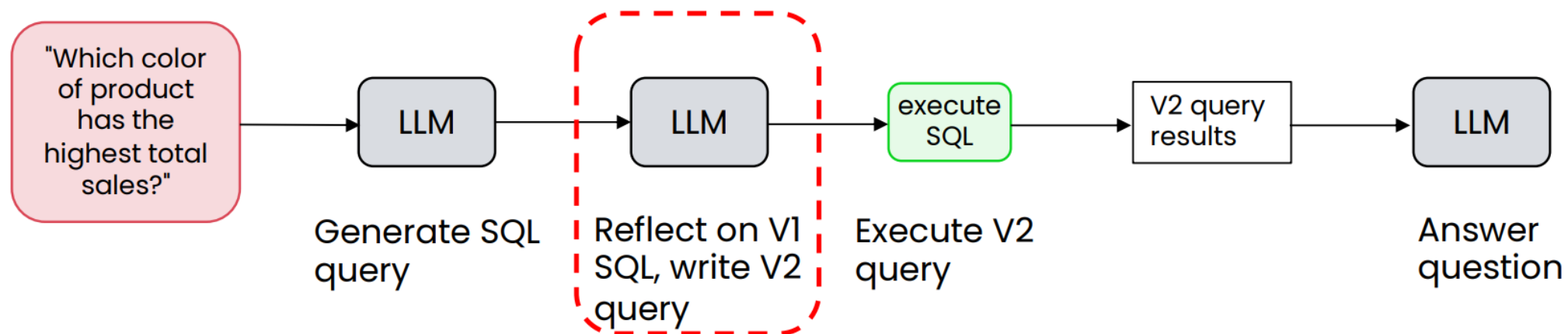


```
import matplotlib.pyplot as plt
import pandas as pd
```

```
# Filter for Q1 sales
q1_sales = df[df['quarter'] == 1]
....
```

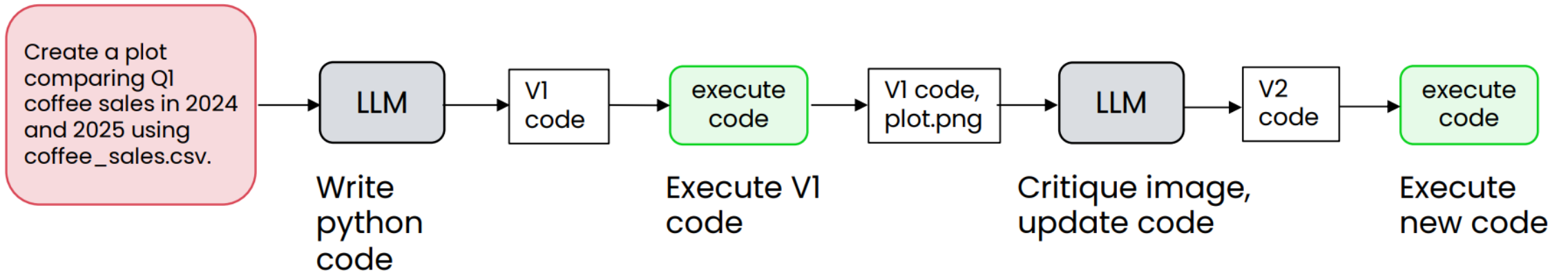


Evaluation

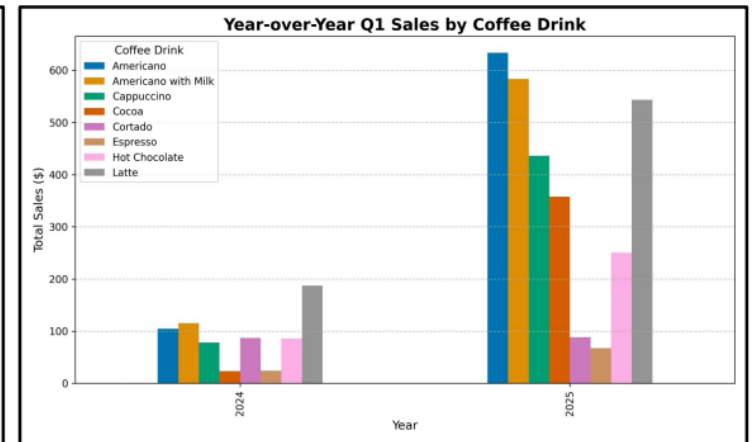
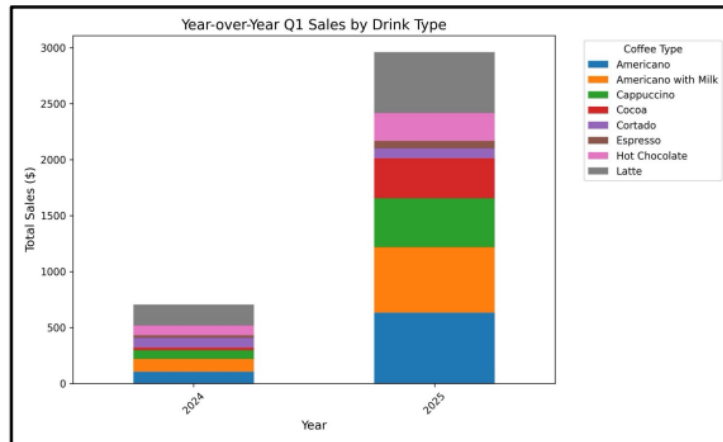


Prompts	Ground truth answer	No reflection	With reflection
Number of items sold in May 2025?	1201	980	1201
Most expensive item?	Airflow sneaker	Airflow sneaker	Airflow sneaker
How many styles carried?	14	14	14

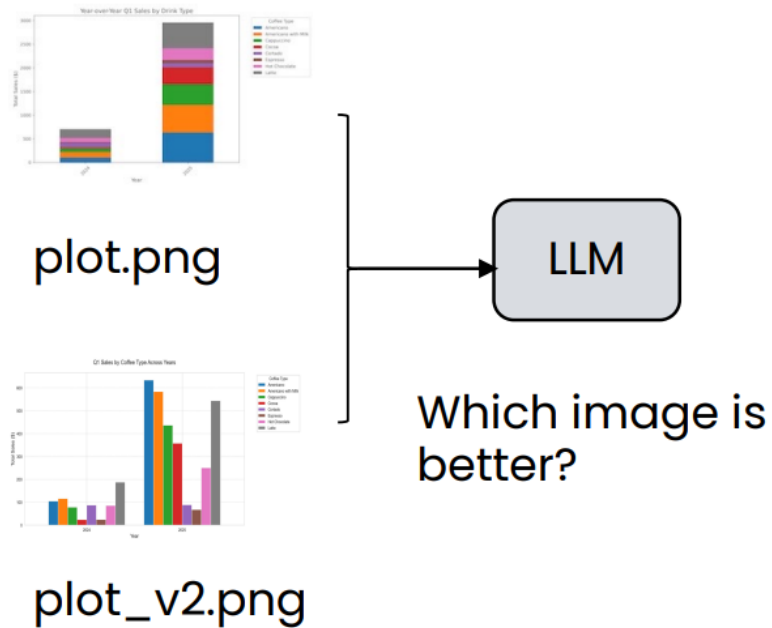
Run each time you change reflection prompt



Can you measure which chart is better with an eval?



Using an LLM as a judge



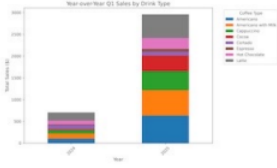
Known issues with using LLMs for comparison:

- Answers often not very good
- Position bias

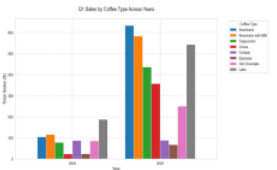
A B

LLM picks A more often

Grading



plot.png



plot_v2.png

LLM

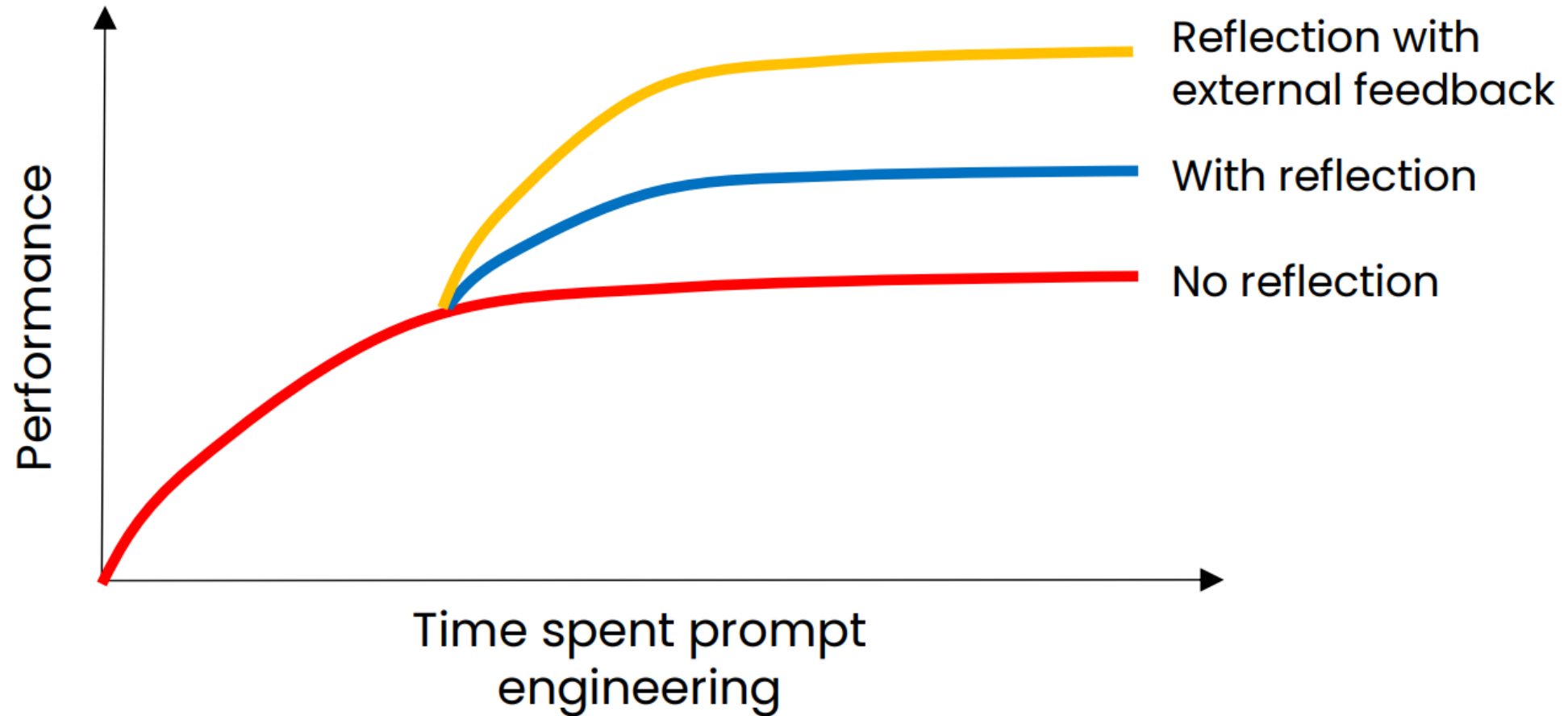
Rubric

Assess the attached image against this quality rubric. Each item should receive a score for 1 (true) or 0 (false). Return the scores for each item as a json object

1. Has clear title
2. Axis labels present
3. Appropriate chart type
4. Axes use appropriate numerical range
5. ...

Input	No reflection	With reflection	Run each time you change reflection prompt
User query 1	4	6	
User query 2	5	8	
....			

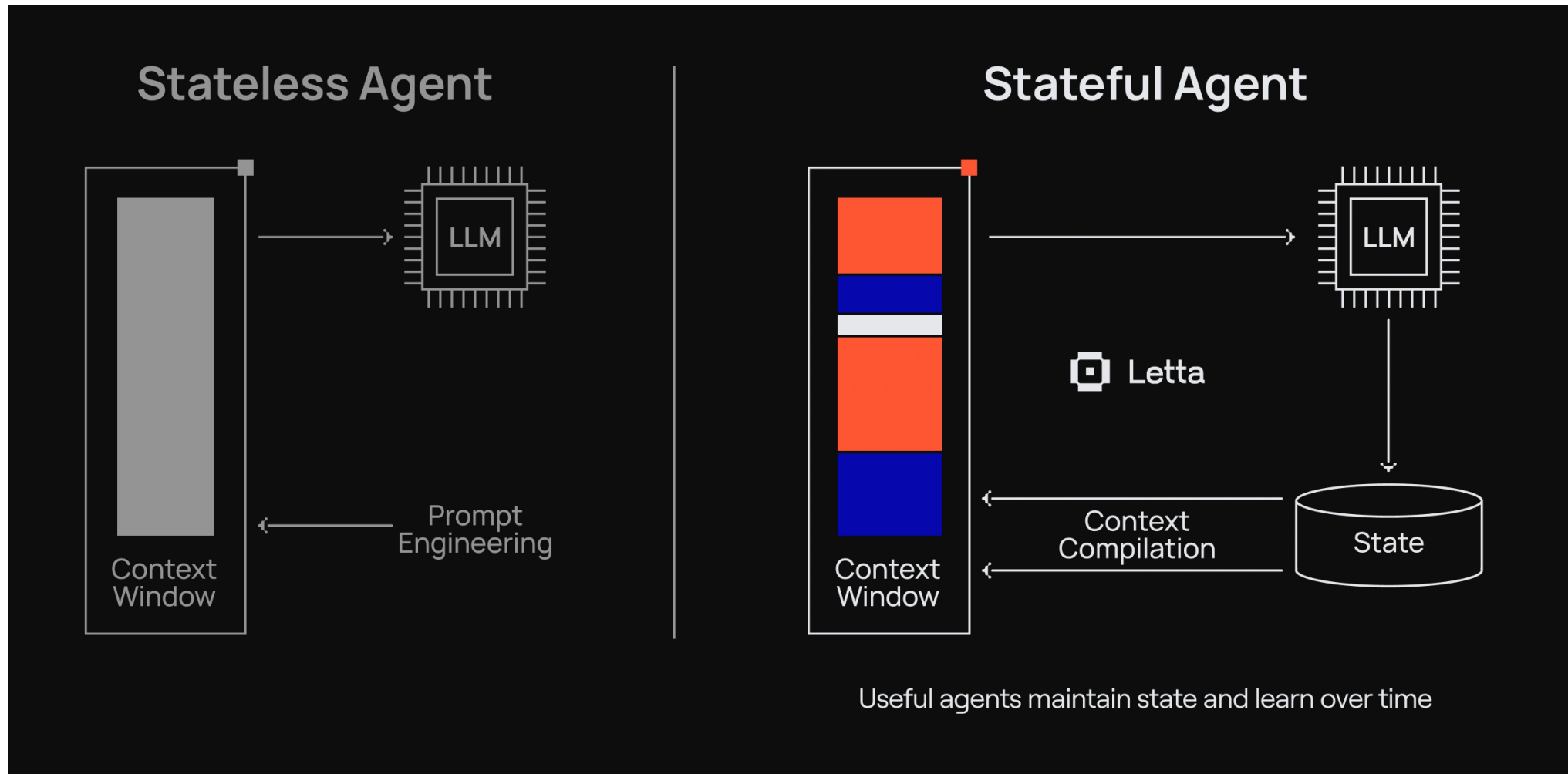
Reflection or prompt engineering?



Agentic design patterns

1. Reflection 
2. Tool use
3. Planning
4. Multi-agent collaboration

LLMs are Stateless



Memory

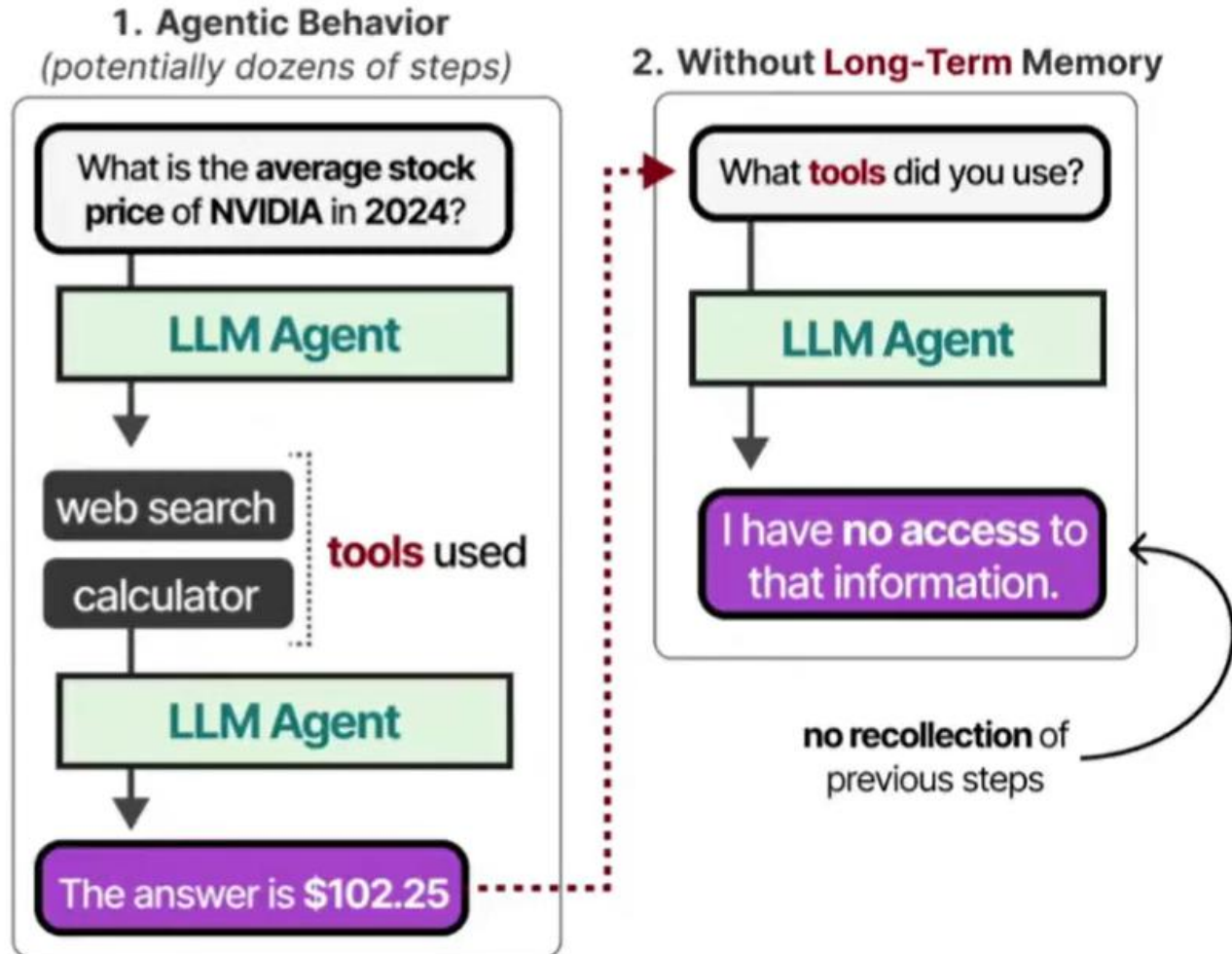
- Short-term memory
- Long-term memory

Without **Short-Term** Memory

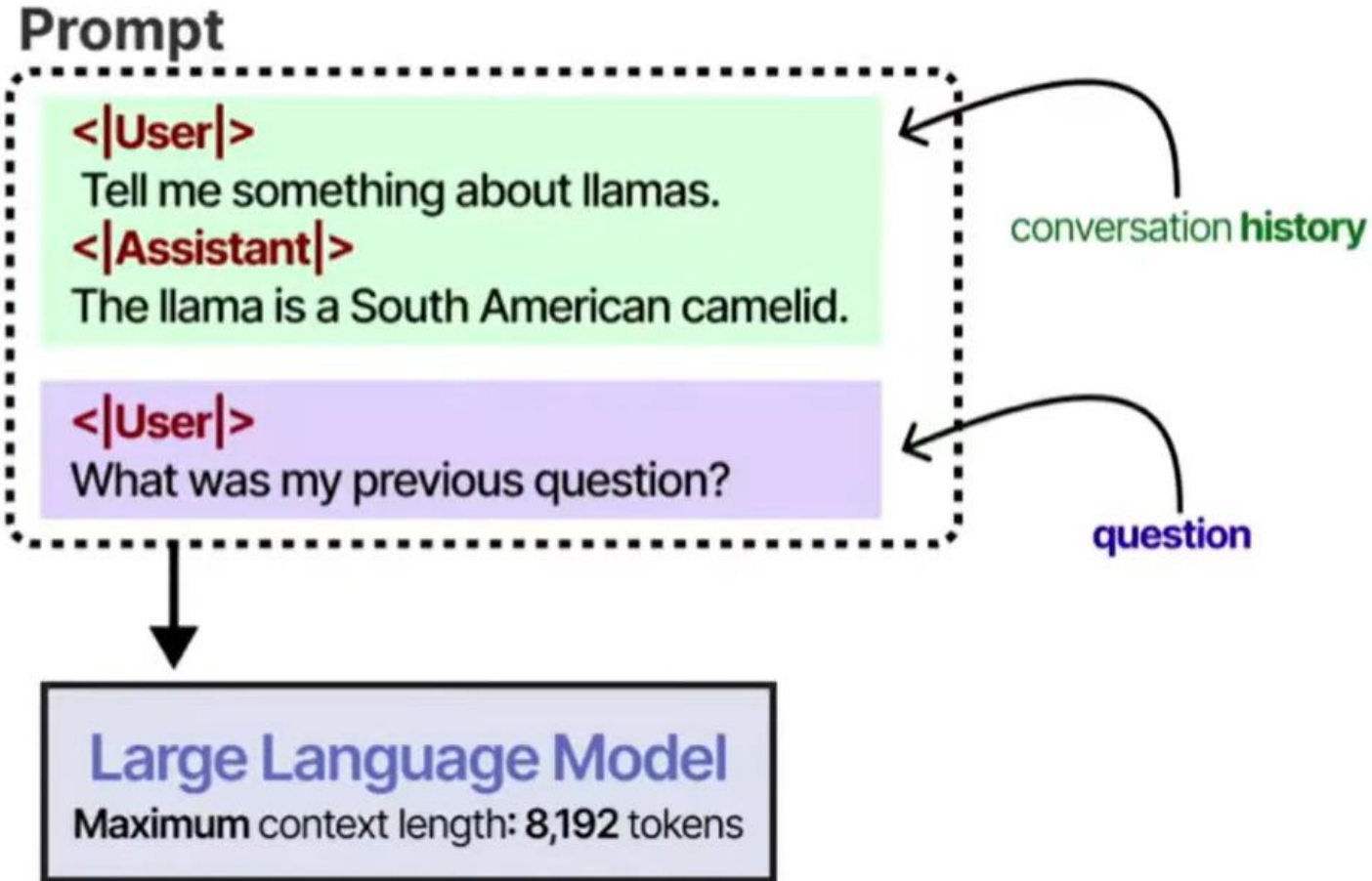


← independent conversations ←

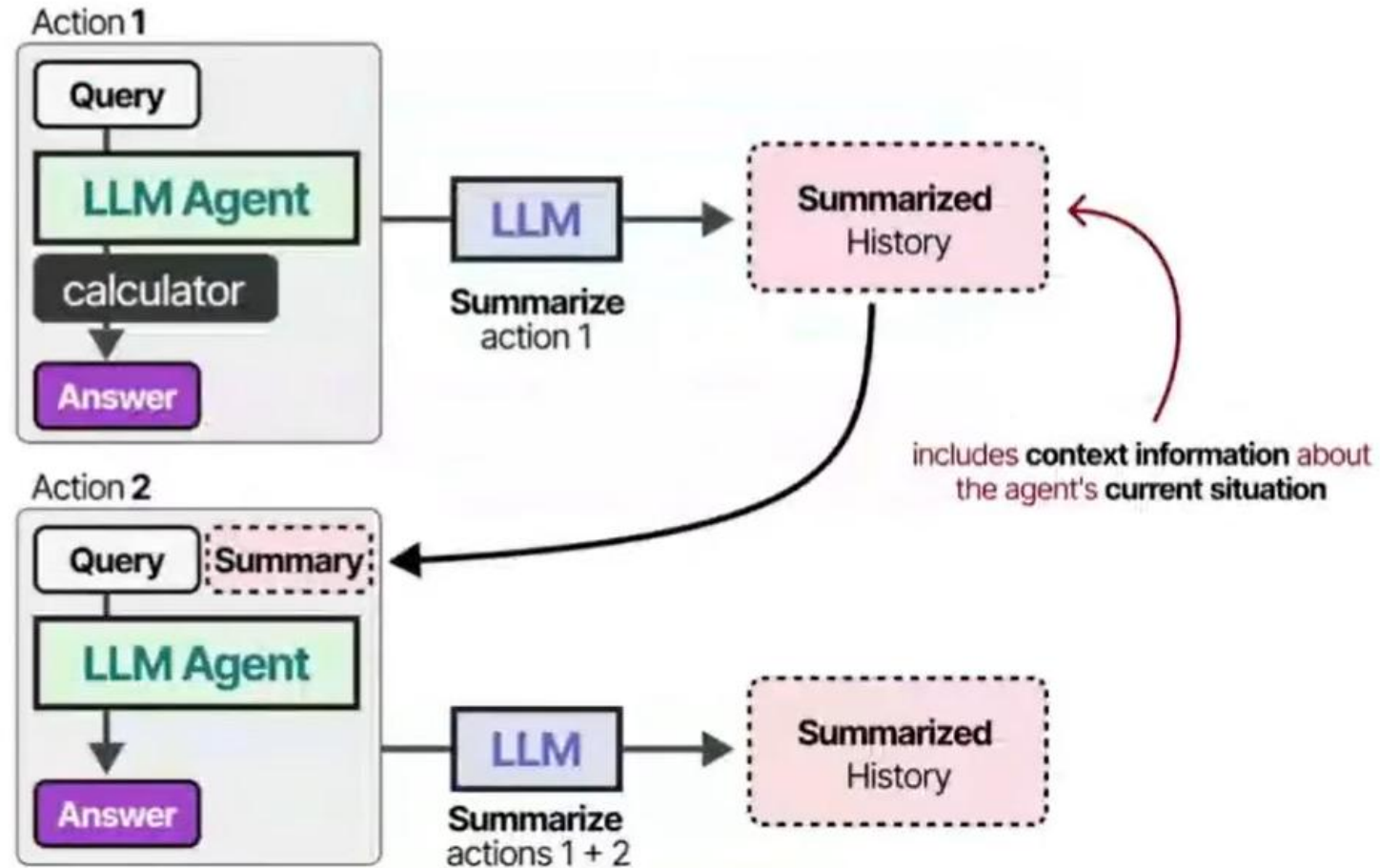
Long-term Memory



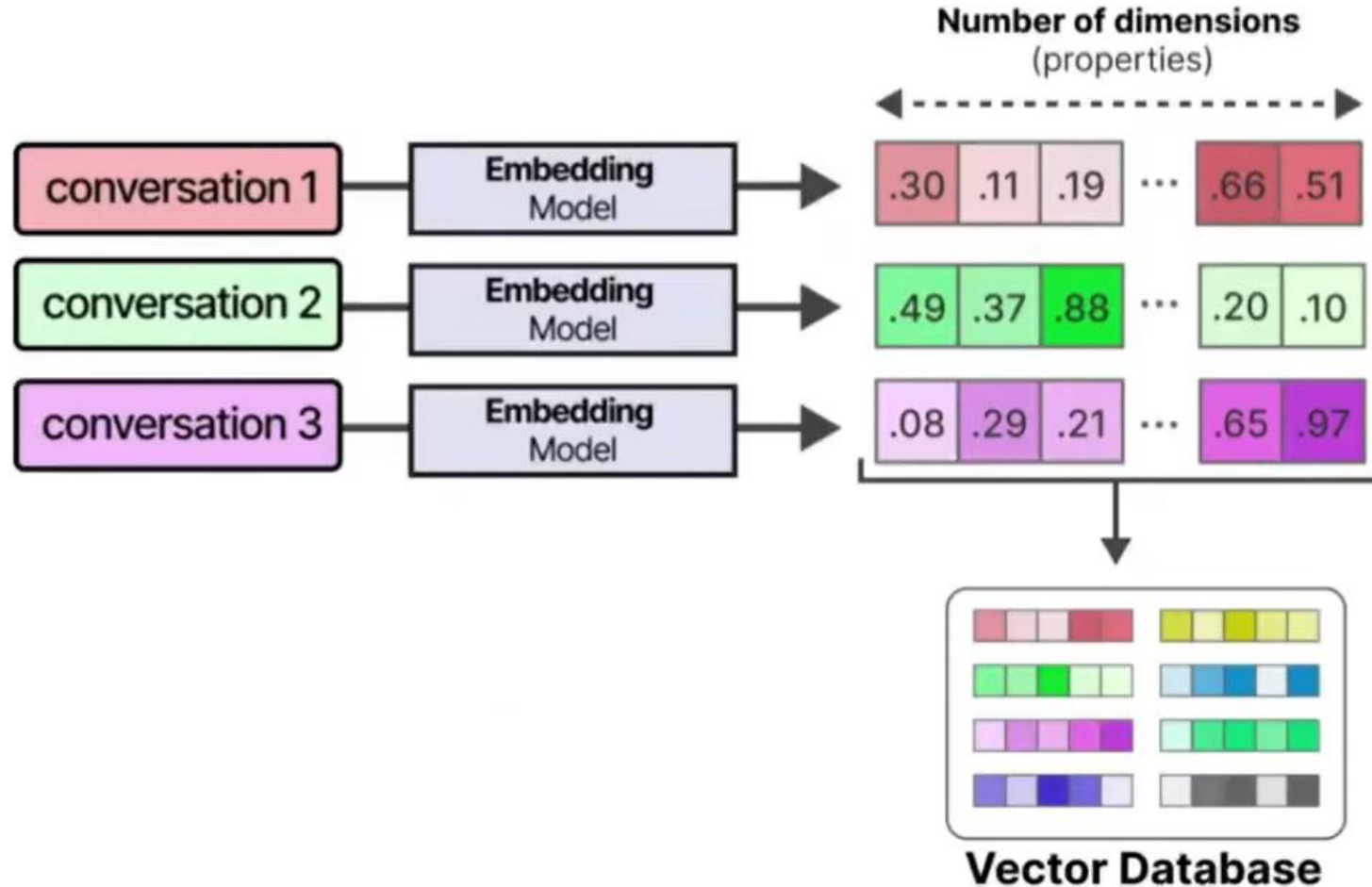
Add Short-term Memory



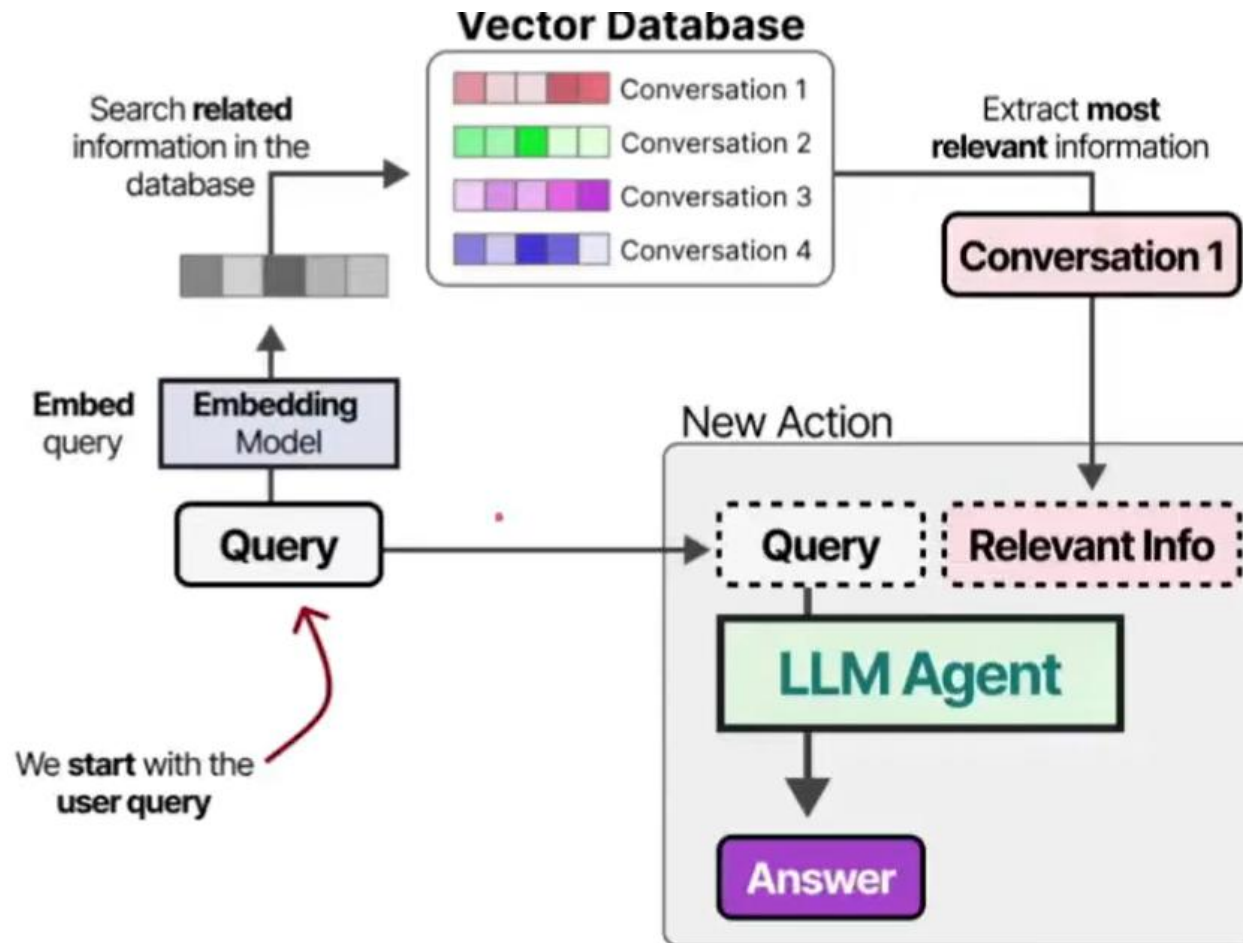
Add Short-term Memory



Add Long-term Memory



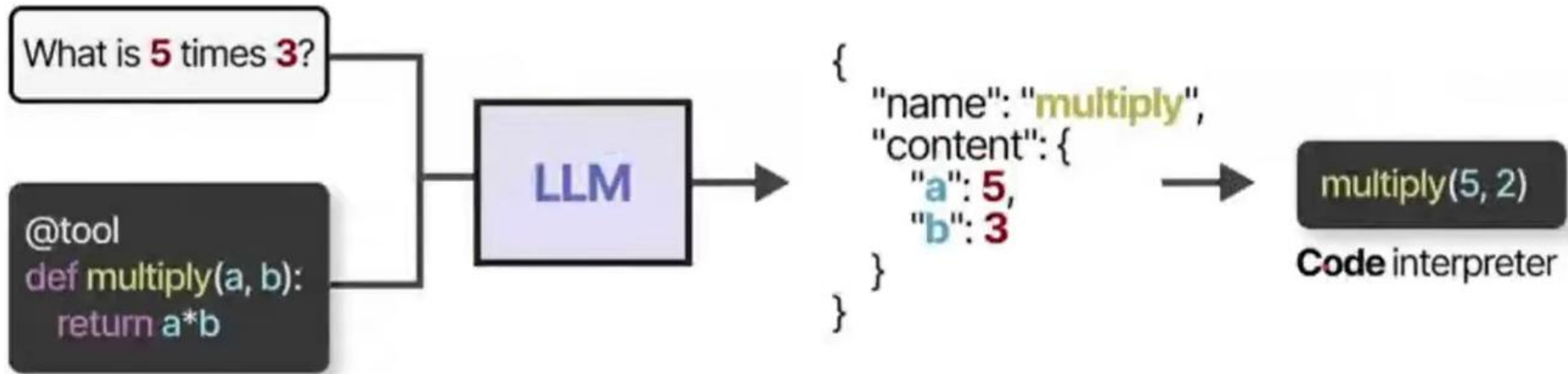
Long-term Memory



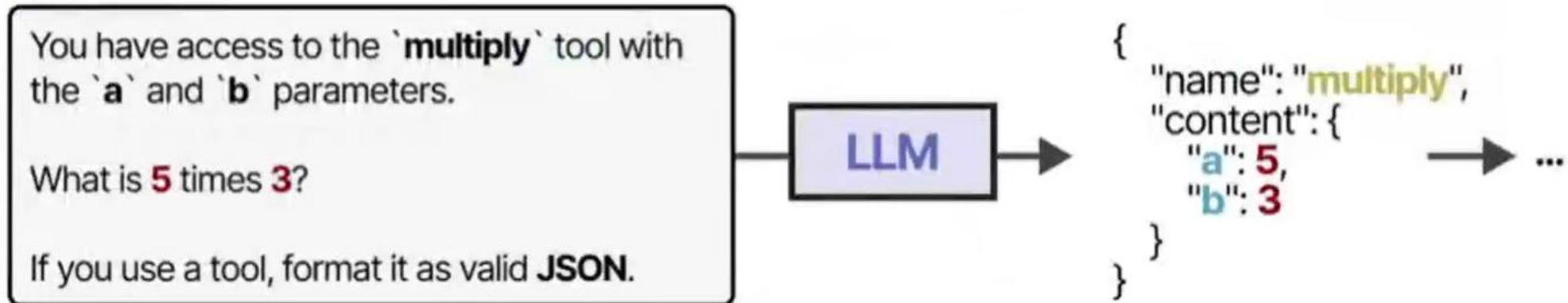
Tools



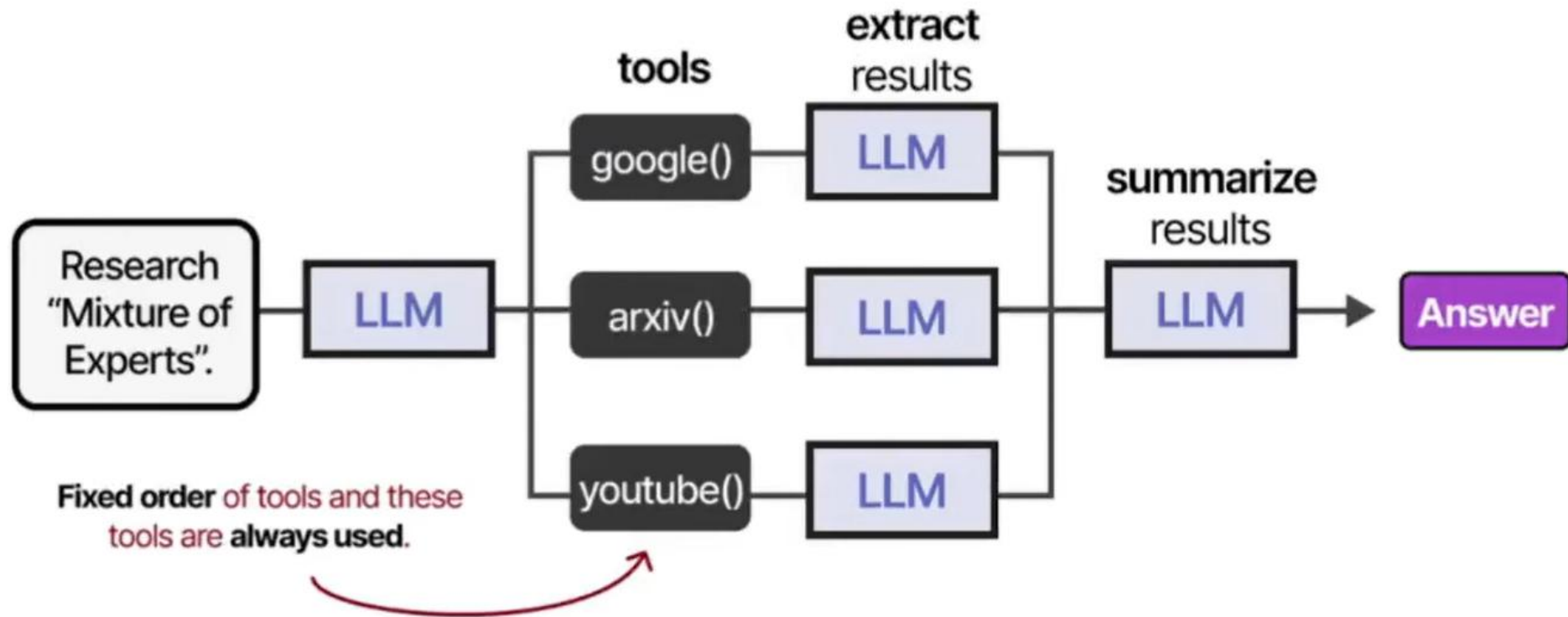
Tools



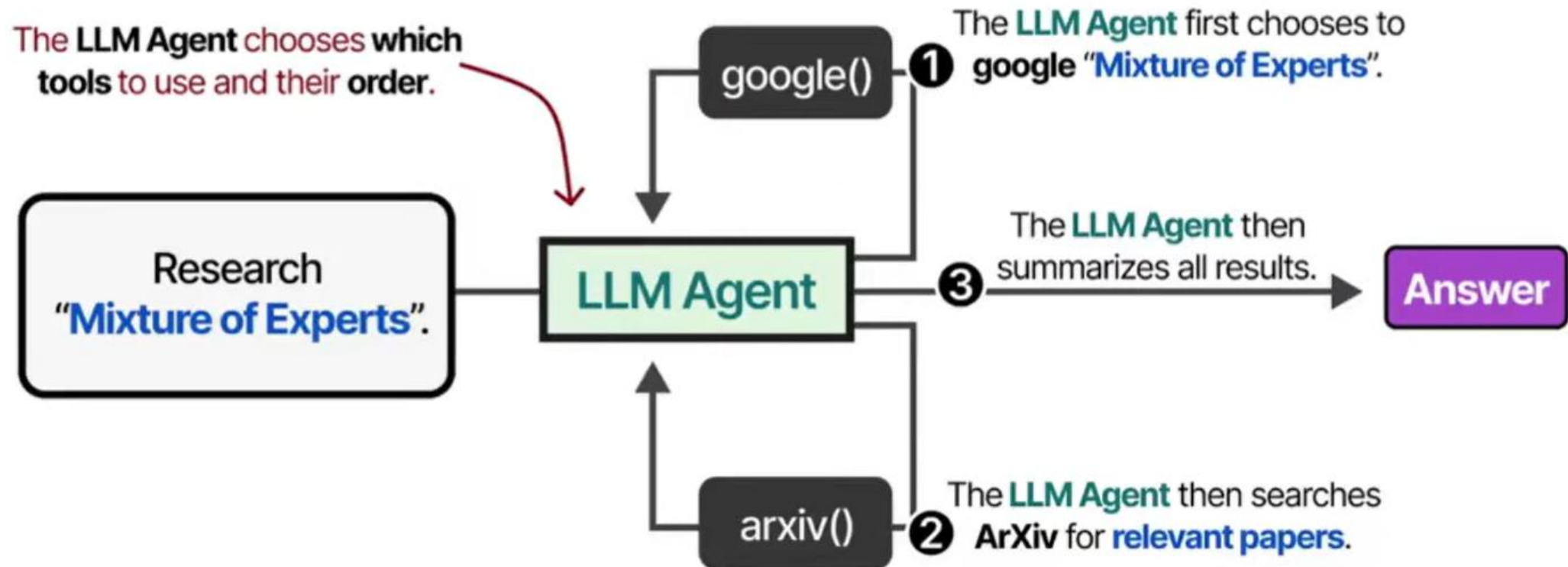
System
prompt



How to call tools? (Fixed Order)

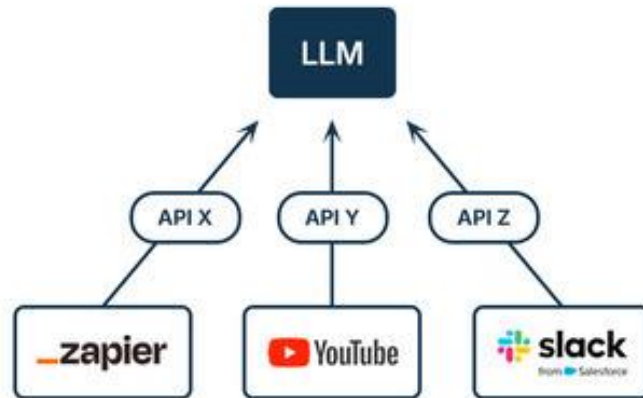


How to call tools? (Reasoning)

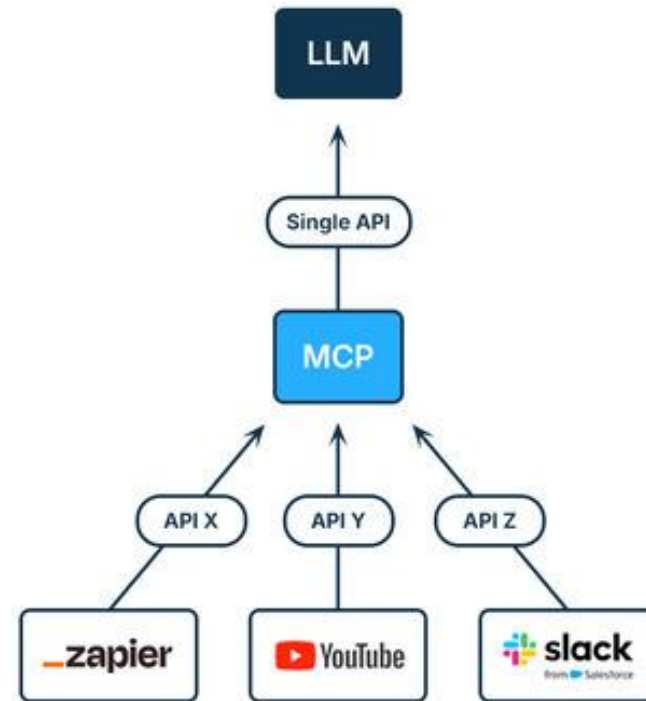


Model Context Protocol (MCP)

Before MCP



After MCP



MCP

Think of MCP like a USB-C port for AI applications. Just as USB-C provides a standardized way to connect electronic devices, MCP provides a standardized way to connect AI applications to external systems.

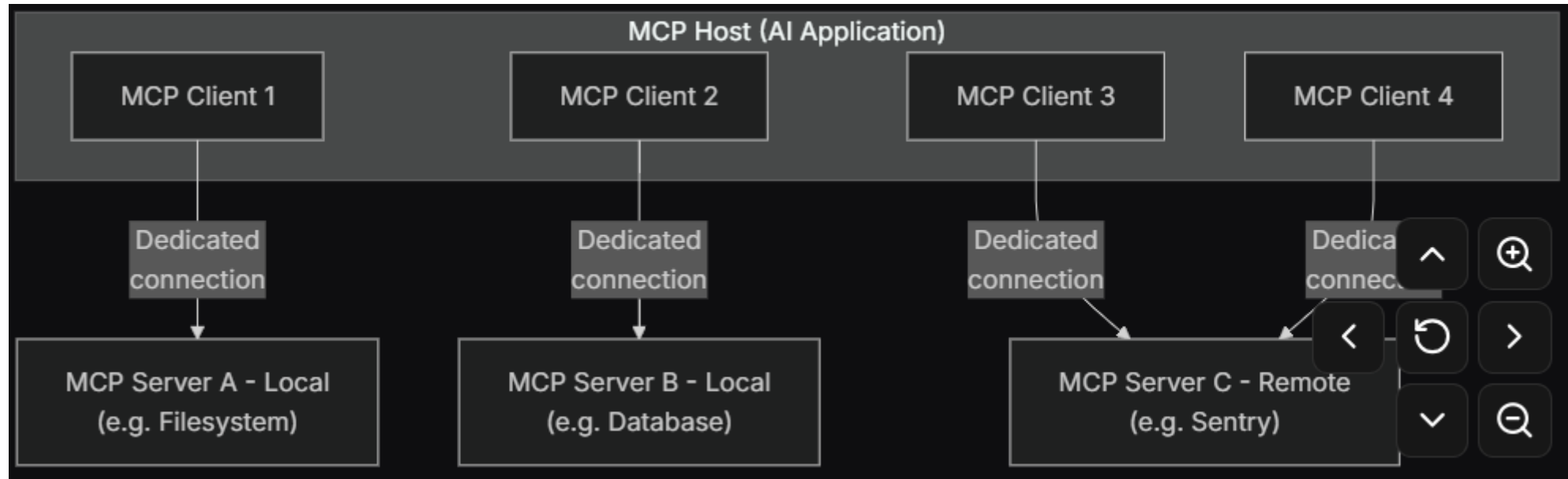
Why does MCP matter?

Depending on where you sit in the ecosystem, MCP can have a range of benefits.

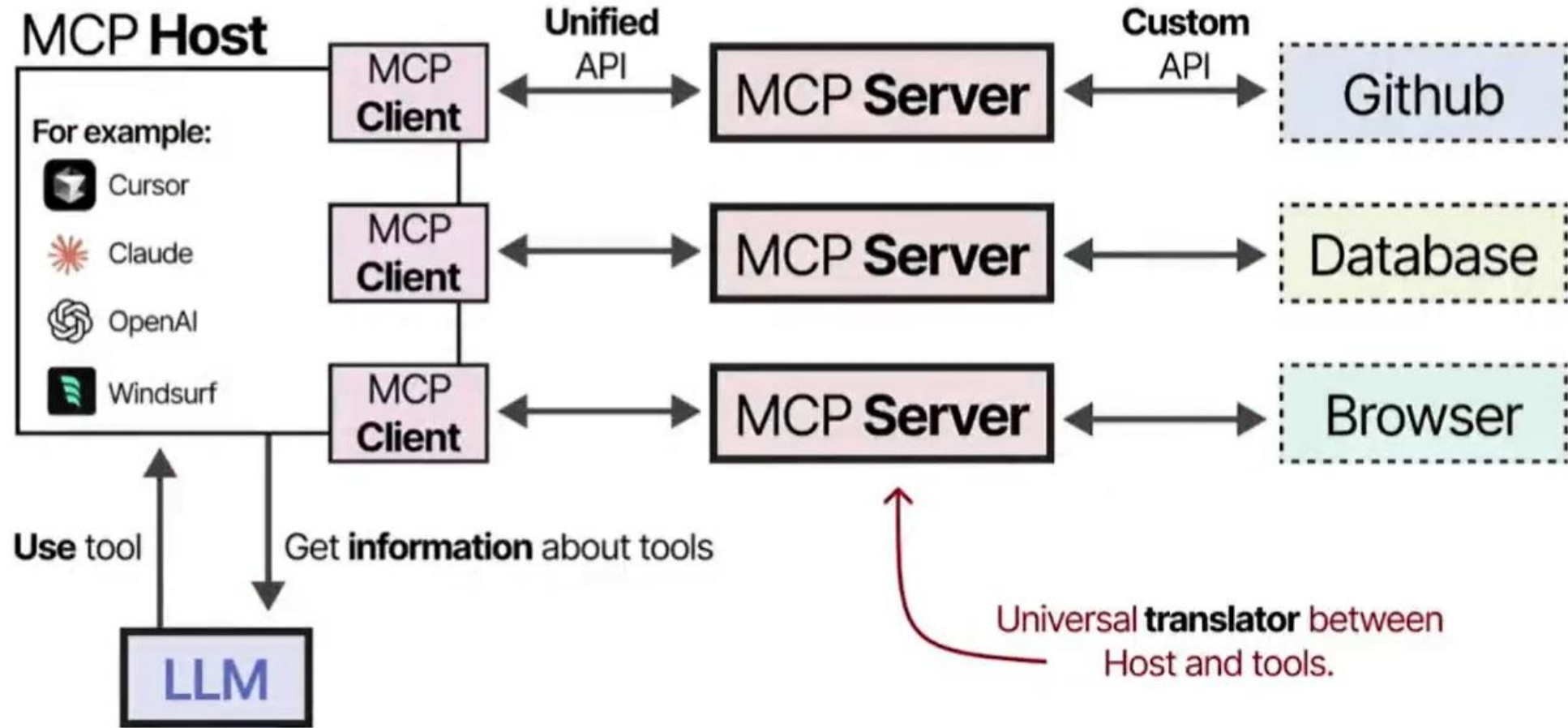
- **Developers:** MCP reduces development time and complexity when building, or integrating with, an AI application or agent.
- **AI applications or agents:** MCP provides access to an ecosystem of data sources, tools and apps which will enhance capabilities and improve the end-user experience.
- **End-users:** MCP results in more capable AI applications or agents which can access your data and take actions on your behalf when necessary.

MCP

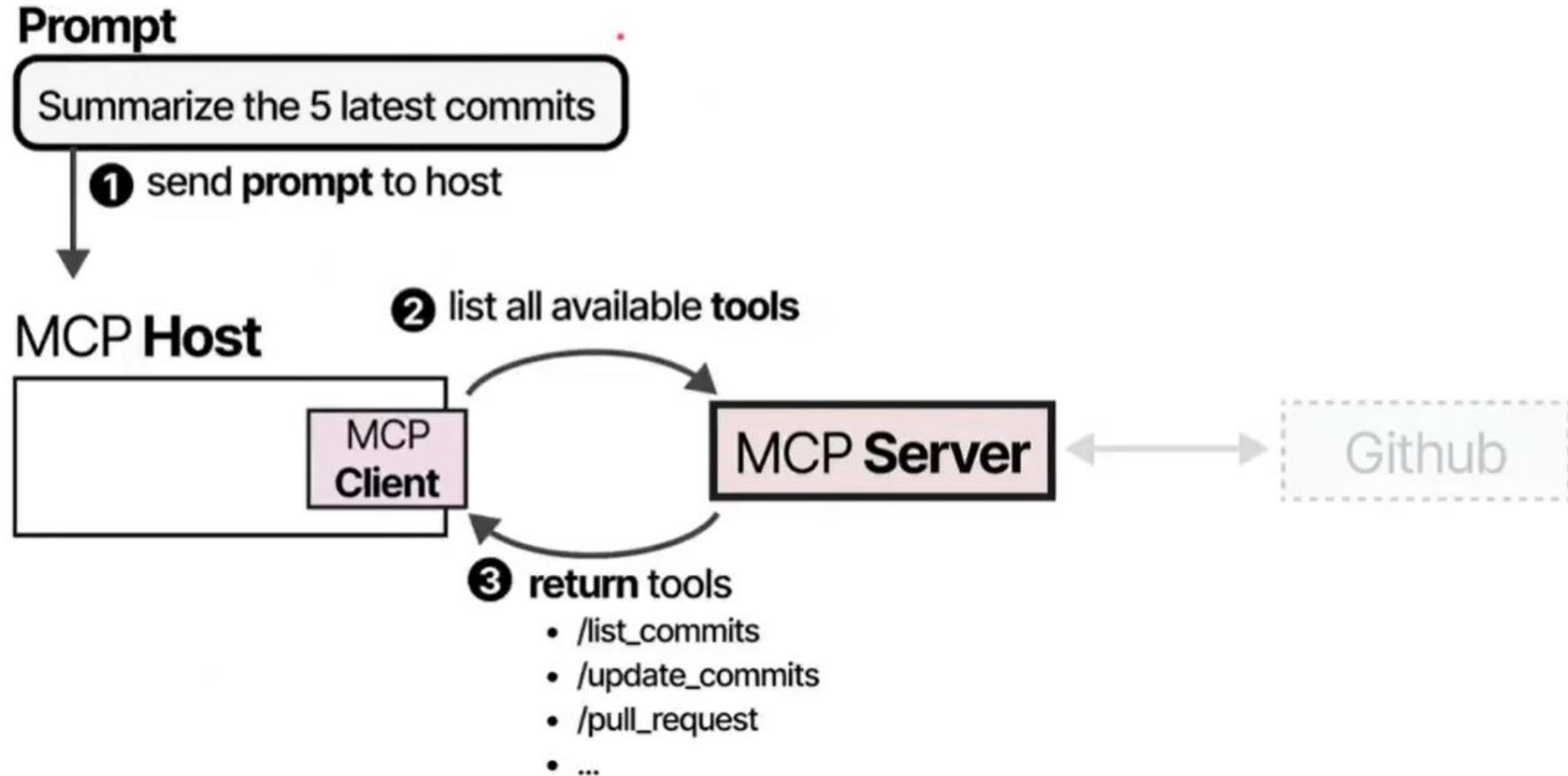
- **MCP Host:** The AI application that coordinates and manages one or multiple MCP clients
- **MCP Client:** A component that maintains a connection to an MCP server and obtains context from an MCP server for the MCP host to use
- **MCP Server:** A program that provides context to MCP clients



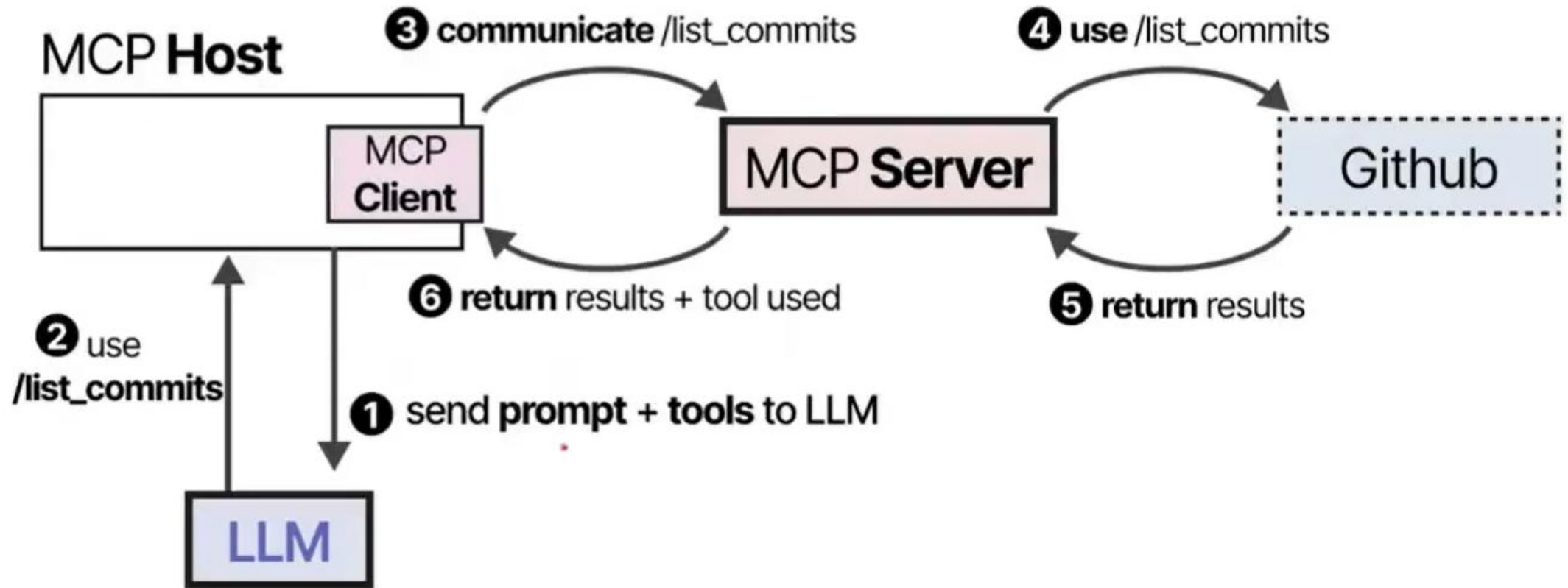
MCP



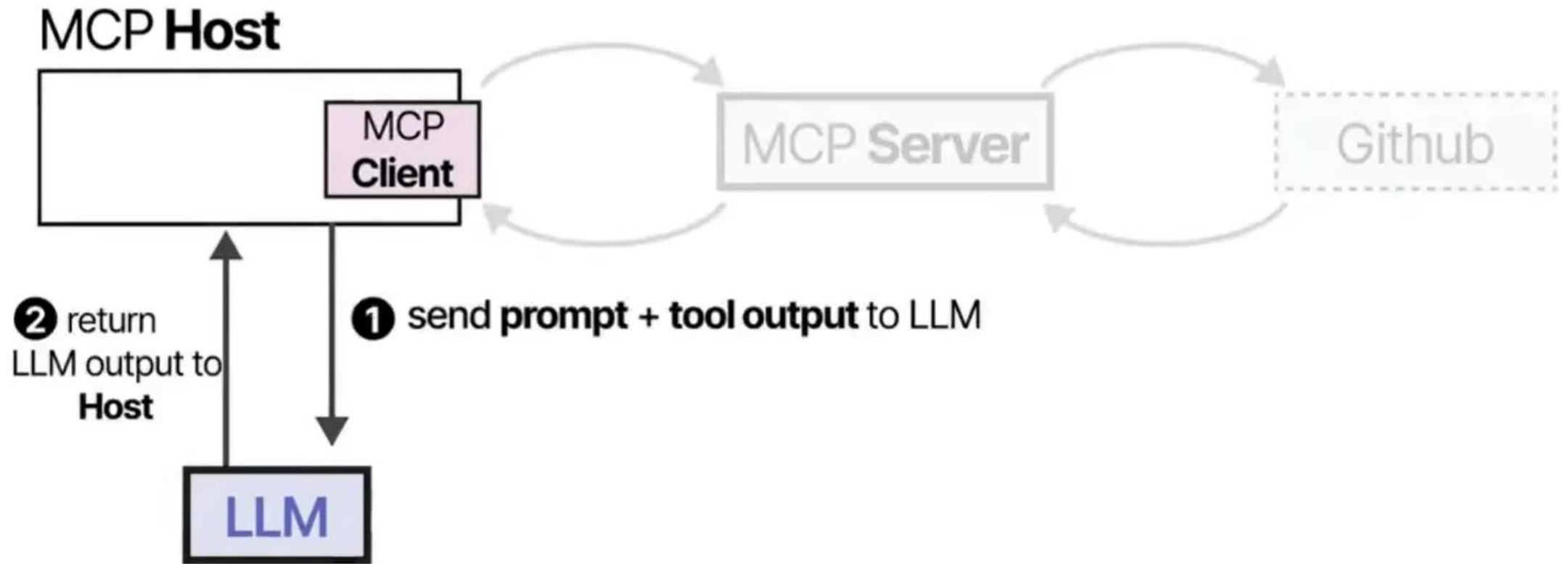
MCP example



MCP example

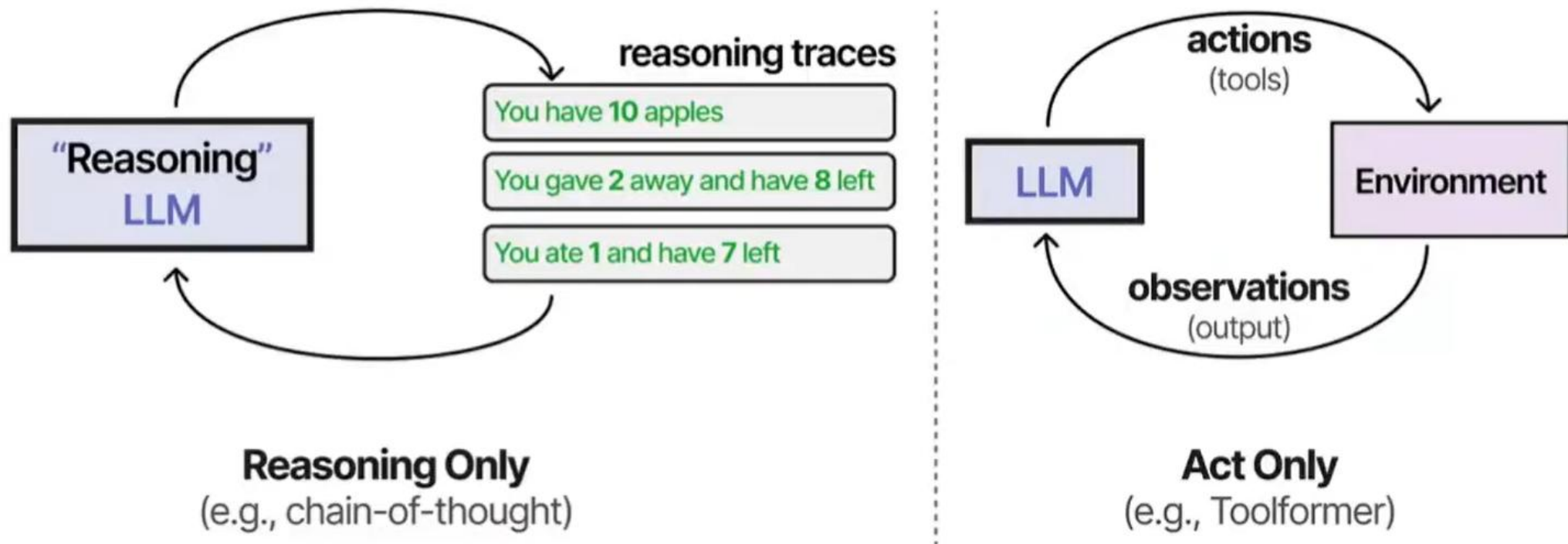


MCP example

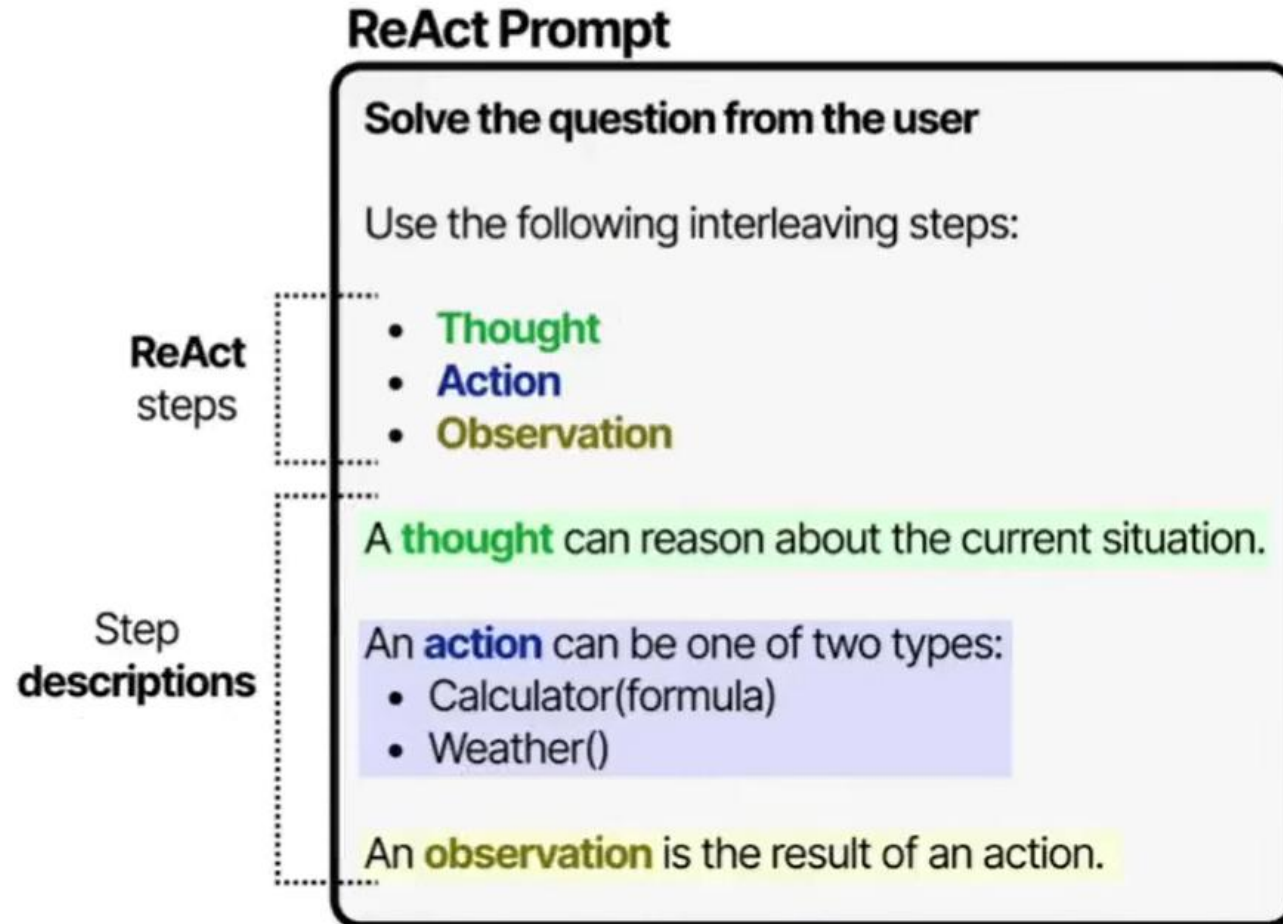


Planning

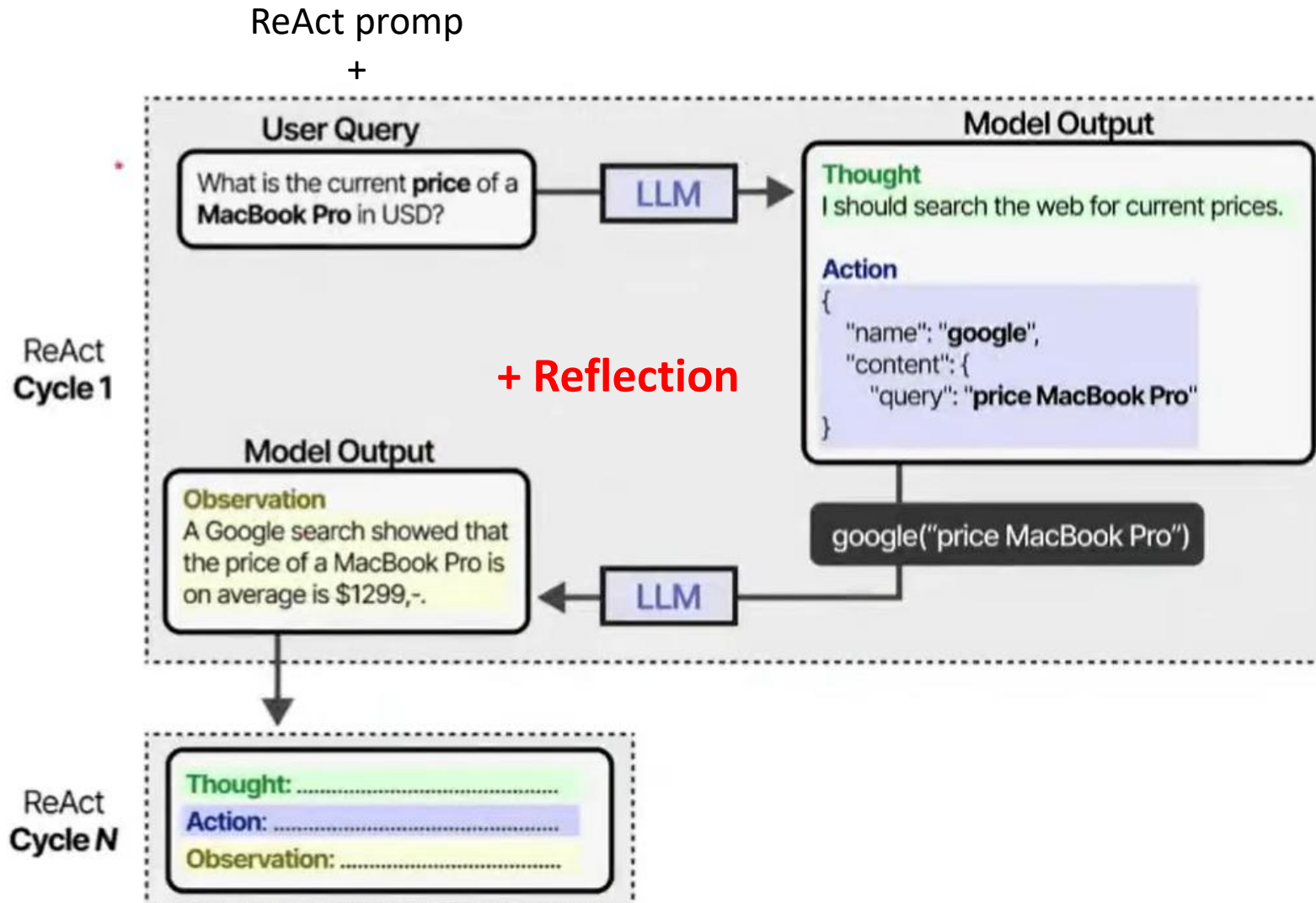
- Reasoning and Acting (ReAct)



ReAct prompt

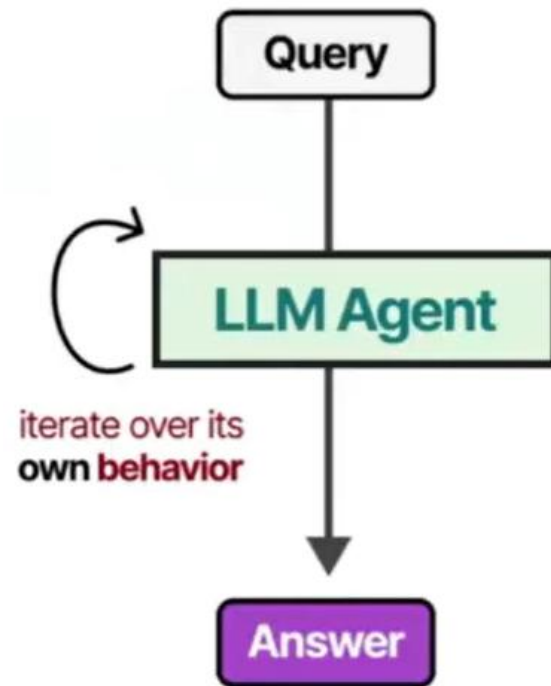


ReAct

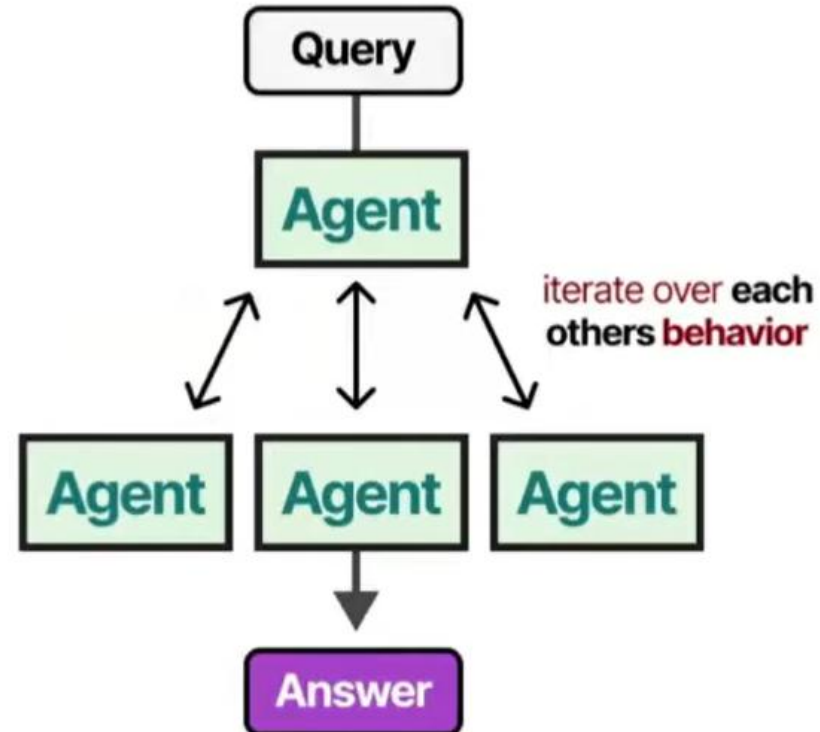


Multi-Agent Systems

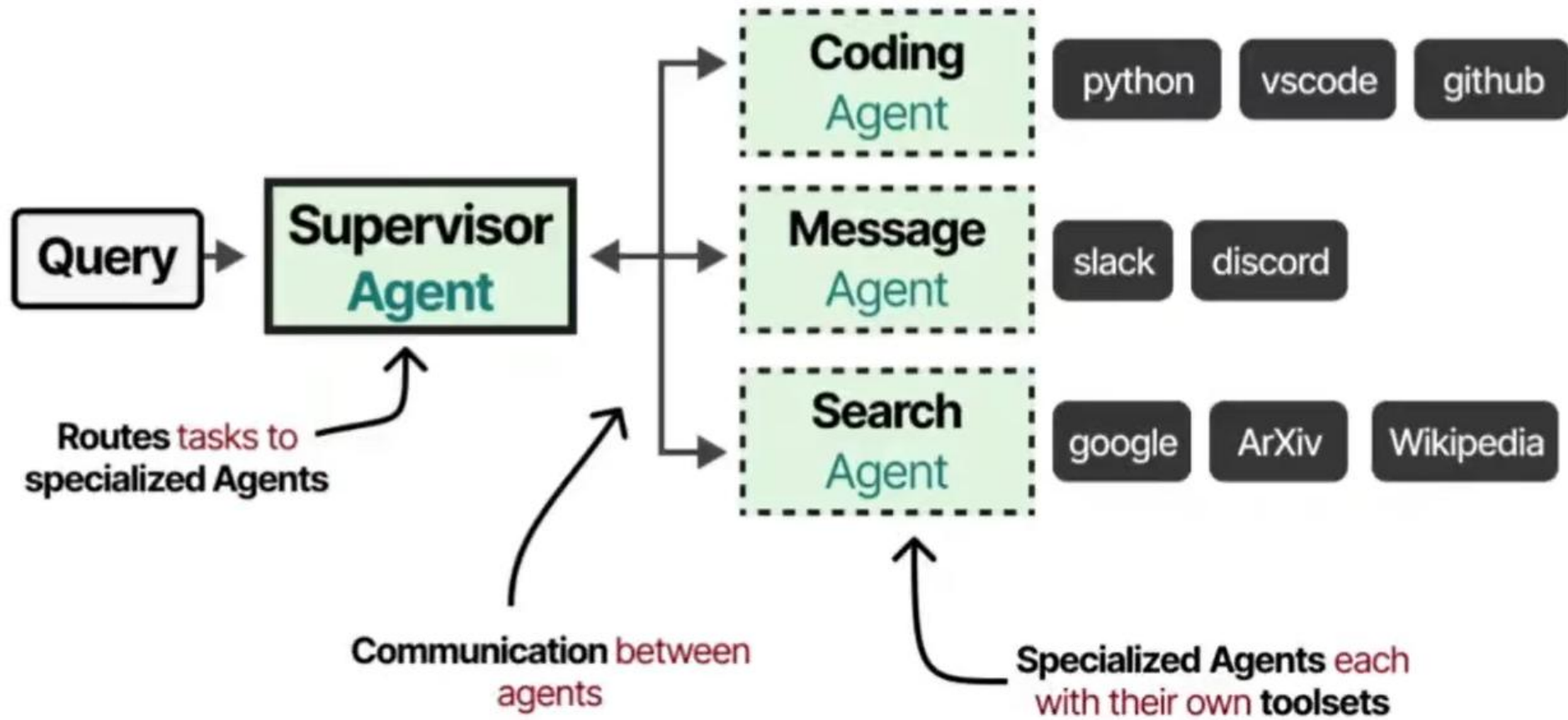
Single Agent



Multi-Agent



Agents in different roles



Nexus framework

